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Real-To-Sim Learning for Dexterous Arm and Hand-Manipulation Tasks

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Authorship Statement

This dissertation is based on the results of research carried out by myself, is my own composition, and has not been previously presented for any other certified or uncertified qualification.

The research was carried out under the supervision of Mr. Paul Pulis.

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Abstract

This dissertation evaluates a real-to-simulation (R2S) pipeline for training a dexterous virtual arm and hand from captured human motion. The prototype uses a physics-based simulation environment with a seven-degree-of-freedom arm and a five-finger hand. Human hand intent is captured through vision-based hand tracking and converted into robot-compatible control values during demonstration recording. Two learning conditions are compared: Condition A uses Behavioural Cloning (BC) from the recorded demonstrations, while Condition B uses early BC assistance followed by Proximal Policy Optimisation (PPO) refinement.

BC-only learning reaches 0% target-region success at the final reported Condition A checkpoint, approximately 1.206M logged steps. BC-assisted PPO achieves approximately 95% target-region success under the fixed training layout. An 800-episode robustness evaluation shows that the refined policy is brittle under spatial variation: success drops to 19% when the cube spawn is offset by 6 cm, and to 23% when the same offset is combined with observation noise. Observation noise alone does not reduce success, with both tested noise-only conditions achieving 100%.

The success metric measures whether the cube enters the target placement radius, not whether it is stably released. The system therefore demonstrates simulated grasp-and-transport-to-target behaviour rather than full autonomous pick-place-release. The findings are descriptive because one training run was completed per condition.

Keywords: dexterous manipulation, real-to-simulation, Behavioural Cloning, Proximal Policy Optimisation, hand tracking, reinforcement learning, spatial perturbation.

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Contents

Authorship Statement i

Copyright Statement ii

Abstract iii

Acknowledgements iv

List of Abbreviations xiv

Symbols xv

1 Introduction 1

1.1 Problem Statement 1
1.2 Background 2
1.3 Motivation 3
1.4 Research Aim, Questions, and Hypotheses 4
1.5 Scope and Limitations 6
1.6 Dissertation Structure 6

2 Literature Review 8

2.1 Overview: The Engineering Challenge of Dexterity 8
2.1.1 Search Index and Source Matrix 9
2.2 Machine Learning Paradigms for Robotic Control 10
2.3 Skill Acquisition via Imitation Learning 12
2.4 The Problem of Information Fragility 13
2.5 Stochastic Policy Refinement via Reinforcement Learning 14
2.5.1 The Hybrid BC→RL Pattern 15
2.6 The Real-to-Simulation Paradigm 17
2.6.1 Capture Fidelity, Demonstration Quality, and Policy Initialisation 17
2.6.2 Kinematic Abstraction and Embodiment Gap 18
2.7 Synthesis of Reviewed Literature 19

2.7.1 Research Gap 20
2.8 Literature Synthesis and Methodological Positioning 21

3 Research Methodology **23**
3.1 Overview 23
3.1.1 Research Aim 24
3.1.2 Research Questions and Hypotheses 24
3.1.3 Research Scope 25
3.1.4 Research Limitations 26
3.2 Research Framework and Formalisation 27
3.2.1 Philosophical Positioning 27
3.2.2 Experimental Design 28
3.2.3 Experimental Variables 28
3.2.4 Problem Formalisation as a Discrete-Time MDP 30
3.3 Implementation 31
3.3.1 Architecture 31
3.3.2 Vision-Based Capture 32
3.3.3 Kinematic Abstraction 33
3.3.4 Unity ML-Agents Training Environment 34
3.3.5 Grasp Validation Logic 37
3.3.6 Reward Function and Reward Scaling 37
3.3.7 Software Testing 39
3.4 Phase I: Pre-Testing and System Verification 39
3.4.1 Functional Implementation Testing 39
3.4.2 Communication and Latency Check 40
3.4.3 Heuristic Verification 40
3.5 Phase II: Pilot Testing and Model Optimisation 41
3.5.1 Initial Learning Performance 41
3.5.2 Hyperparameter Selection and Observations from Pilot Runs . 41
3.5.3 Reward Gating Verification 44
3.6 Phase III: Implemented Testing Matrix 44
3.6.1 A/B Training Comparison 45
3.6.2 Robustness Evaluation: Eight-Condition Stress Test 46
3.7 Evaluation Framework and Statistical Analysis 49
3.7.1 Performance Metrics 49
3.7.2 Proxy Grasp Reliability Analysis 50
3.7.3 Descriptive Analysis Protocol 50
3.8 Ethical and Validity Constraints 51
3.8.1 Ethical Data Protocol 51

3.8.2 Internal and External Validity 51
3.9 Chapter Summary 52

4 Analysis of Results and Discussion 53
4.1 Chapter Overview 53
4.2 Relationship to the Implemented Evaluation Matrix 54
4.3 Training Data and Demonstration Boundary 55
4.4 Condition A: Behavioural Cloning Only 56
4.5 Condition B: BC-Assisted PPO Refinement 58
4.6 A/B Comparison and Refinement Gain 59
4.7 Robustness Evaluation of the Final Condition B Policy 63
4.8 Interpretation of Spatial Robustness 65
4.9 Observation Noise and Combined Perturbation 66
4.10 Efficiency and Time-to-Success 67
4.11 Comparison with Related Work 68
4.12 Statistical and Replication Boundary 72
4.13 Discussion Against the Research Questions 73
4.13.1 RQ1: Can the virtual arm and hand accomplish the manipulation sequence? 73
4.13.2 RQ2: Does the learned policy reproduce the task-level structure of demonstrations? 74
4.13.3 RQ3: Is the system efficient and near-real-time? 75
4.14 Main Limitations Identified by the Results 76
4.14.1 Proxy Interpretation of Grasp-Validation Outcomes 76
4.14.2 Proxy Precision, Recall, and Operating Curve Across Test Conditions 78
4.15 Chapter Summary 81

5 Conclusions and Recommendations 83
5.1 Chapter Overview 83
5.2 Summary of Findings 84
5.3 Answers to the Research Questions 85
5.3.1 RQ1: Can a virtual arm and hand be trained within a simulation environment with R2S learning using captured human motion to accomplish basic grasping, lifting, transporting, and target-region completion? 85
5.3.2 RQ2: To what extent can the virtual arm and hand reproduce the task-level structure of the demonstrations and manipulate objects within the simulator? 86

5.3.3 RQ3: How efficient is the learned system with respect to performance, training time, and real-time or near-real-time use within the environment? . . . 87
5.4 Hypothesis Evaluation . . . 87
5.5 Contribution of the Dissertation . . . 88
5.6 Limitations . . . 89
5.7 Recommendations and Future Work . . . 90
5.8 Final Remarks . . . 91

Bibliography **93**

A Implementation and Reproducibility Artefacts **98**
A.1 Project Artefacts . . . 98
A.2 Agent Interface Summary . . . 98
A.3 Demonstration Statistics Log . . . 99
A.4 Result Artefact Register . . . 99

B Literature Review Support Matrix **100**
B.1 Peripheral Literature Strands . . . 100
B.2 Reviewed Study Matrix . . . 101

List of Figures

2.1 Literature synthesis route used to organise the review. 9
2.2 BC and RL address different failure modes. BC supplies the behavioural prior; PPO refinement learns corrective behaviour through environmental interaction. 12
2.3 Research gap identified from the reviewed literature. 21

3.1 Research onion framing for the dissertation methodology, adapted from Saunders et al.'s layered research-design model [1]. Each concentric layer represents a methodological commitment, from the broadest philosophical stance (outer) to the data collection and analysis approach (core). 27
3.2 Implemented R2S pipeline: human capture feeds the Unity R2S interface, which records ML-Agents demonstration files used for training under Condition A (BC-only) and Condition B (BC-assisted PPO refinement); the trained checkpoints are then analysed using task-level metrics from training logs and, for Condition B, inference-mode robustness tests. 32
3.3 MediaPipe hand tracking overlay showing 21 landmarks on the operator's hand. The scalar flexion values for thumb, index, middle, ring, pinky, and spread are derived from landmark-based bend angles and angular separation measures before being transmitted to Unity. . . . 33
3.4 Unity simulation environment. The KUKA iiwa7 arm and SCHUNK SVH hand are shown with the cube at the fixed spawn location (left) and the target marker (right, green radius). The hand is in an open pose before demonstration recording. 35

3.5 Stress-test condition overview. **(a)** Top-down view of the cube-spawn perturbation zones; each nested square is the uniform random draw region for T1–T5 (scale: 1 unit = 6 cm). The orange dashed boundary marks T3 (±6 cm), the medium spatial perturbation condition. **(b)** Observation noise profiles for T6–T8: per-step Gaussian noise $\mathcal{N}(0, \sigma^2)$ is added independently to all continuous observations (binary flags are excluded). T8 combines the T3 spatial offset with low noise. 47

4.1 Comparison of key converged training metrics for Condition A and Condition B. Both conditions reached a closure-validator rate of 100%, indicating that hand closure was learned under both regimes. The critical separation is in post-closure outcomes: Condition A produced drops in approximately 70% of logged episodes and 0% target-region success, while Condition B reduced the drop rate to approximately 10% and reached approximately 95% target-region success. Drop rate is an approximate percentage derived from the ML-Agents 10-episode logging window at convergence; closure-validator rate and target-region success are taken from the final training checkpoint. 62

4.2 Condition B stress-test success rates. The final policy achieves 100% success under fixed-layout and observation-noise-only conditions, but fails sharply under spatial spawn perturbation. 64

4.3 Episode outcome breakdown across all eight stress-test conditions. Each bar sums to 100%. Success dominates under fixed-layout and noise-only conditions; drop rate rises sharply with spatial perturbation while timeout remains comparatively low. 65

4.4 Success rate as a function of cube-spawn offset radius for noise-free conditions (T1–T5). T8 (combined ±6 cm with low observation noise) is shown as a separate marker for comparison. Performance collapses sharply beyond ±3 cm, and the addition of noise at the same offset does not materially worsen the outcome. 66

4.5 Mean minimum distance of the object to the target region across all stress-test conditions. Conditions T1, T6, and T7 achieve proximity near the 0.15 m success threshold. Under large spatial offsets (T3–T5, T8) the object frequently fails to approach the target region at all, confirming that failure is not a near-miss but a full task breakdown. . 68

4.6 Proxy precision operating curve across all eight stress-test conditions. Recall is approximately 1.00 in every condition because the closure-validator fired in every episode; this curve therefore summarises the proxy operating profile across varying task difficulty. Conditions T1, T6, and T7 achieve precision 1.00. Spatial perturbation degrades precision sharply, with T4 (± 10 cm) the worst-performing condition at 0.05. T5 recovers slightly due to edge-boundary artefacts at very large offsets. 80

List of Tables

2.1 Search index and source-cluster matrix for the literature review. . . . 10
2.2 Representative BC-only versus hybrid BC→RL results from reviewed studies. . . . 15
2.3 Capture-method trade-off across the reviewed literature. The dissertation uses an intermediate Kinect–MediaPipe position: more instrumented than a basic webcam, less costly than MoCap or dedicated teleoperation. . . . 17

3.1 Experimental variable structure used in the final implementation. . . 30
3.2 Final observation-space composition. . . . 36
3.3 Final continuous action-space composition. . . . 36
3.4 Main reward components used during PPO refinement (Condition B). Values are as implemented in `DexterousPickPlaceAgent.cs`. . . . 38
3.5 Final training settings used in the implemented learning regimes. PPO-specific parameters (clip ϵ, discount γ, time horizon) apply to the PPO optimisation in Condition B (BC-assisted PPO). Condition A used ML-Agents Behavioural Cloning from the recorded demonstrations only; no extrinsic task reward contributed to the Condition A learning signal. 43
3.6 A/B training comparison matrix. . . . 45
3.7 Eight-condition robustness evaluation matrix. All conditions use the fixed Condition B model. . . . 48

4.1 Results chapter alignment with the implemented evaluation matrix. . 55
4.2 Condition A training outcome summary. . . . 57
4.3 Condition B training and fixed-layout inference summary. . . . 59
4.4 High-level comparison of the completed learning pipelines. . . . 61
4.5 Condition B robustness stress-test results ($n = 100$ episodes per condition). Drop = episode ended without success and without timeout. Time-to-success is computed only over successful episodes. . . . 63
4.6 Relationship between this dissertation and representative empirical literature reviewed in Chapter 2. . . . 70

4.7 Statistical interpretation boundary for the completed results. 72
4.8 Proxy true-positive-like, false-positive-like, true-negative-like, and false-negative-like interpretation categories for GraspValidator outcomes. Categories are descriptive only; no ground-truth grasp labels were independently collected. 77
4.9 Proxy precision, recall, and F1 score per stress-test condition (100 episodes each), derived from the closure-validator proxy framing. Recall is approximately 1.00 in all conditions because the validator fired in every episode. Precision tracks the per-condition success rate and degrades with perturbation severity. 79

A.1 Main repository artefacts supporting the dissertation. 98
A.2 Final ML-Agents interface used by the implemented prototype. 98
A.3 Result artefacts retained with the submitted project for reproducibility checking. 99

B.1 Peripheral literature strands and reason for appendix placement. . . . 100
B.2 Extended reviewed-study matrix supporting Chapter 2. 101

List of Abbreviations

Abbreviation	Meaning
A/B	Two-condition comparison between Condition A and Condition B
BC	Behavioural Cloning
BC→PPO	Behavioural Cloning followed by Proximal Policy Optimisation refinement
BC→RL	Behavioural Cloning followed by Reinforcement Learning refinement
CSV	Comma-Separated Values
DIP	Distal Interphalangeal joint
F1	F1 score; harmonic mean of precision and recall
FPR	False Positive Rate
LTS	Long-Term Support
MCP	Metacarpophalangeal joint
MDP	Markov Decision Process
ML	Machine Learning
ML-Agents	Unity Machine Learning Agents Toolkit
MoCap	Motion Capture
PIP	Proximal Interphalangeal joint
PPO	Proximal Policy Optimisation
R2S	Real-to-Simulation
RGB	Red, Green, Blue image data
RL	Reinforcement Learning
ROC	Receiver Operating Characteristic
RQ	Research Question
SVH	SCHUNK Servo-electric Five-finger Hand
TPR	True Positive Rate
TTS	Time-to-Success
UDP	User Datagram Protocol

Symbols

Symbol	Meaning
General learning notation	
$\mathcal{D}$	Demonstration dataset recorded from the human operator.
S	State or observation space.
A	Action space.
T	Transition function.
R	Reward function.
ρ_0	Initial-state distribution.
t	Discrete timestep within an episode.
s_t	Environment state at timestep t.
a_t	Action selected at timestep t.
s_i	State sample from the demonstration dataset.
a_i^*	Expert or demonstrated action paired with s_i.
a^*	Expert or demonstrated action used in the BC objective.
γ	Discount factor for future rewards.
π	Policy mapping observations or states to actions.
π_θ	Trainable policy parameterised by θ.
θ	Trainable policy parameters.
θ_{old}	Policy parameters before a PPO update.
$\hat{A}_t$	Estimated advantage at timestep t.
$\mathbb{E}$	Expectation operator.
PPO and behavioural cloning notation	
$r_t(\theta)$	PPO probability ratio between current and old policy.
ϵ	PPO clipping parameter.
$\mathcal{L}_{\text{BC}}$	Behavioural Cloning imitation loss.
$\mathcal{L}^{\text{CLIP}}$	PPO clipped surrogate objective.
Evaluation notation	
d_t	Cube-to-target distance at timestep t.
d_{t-1}	Cube-to-target distance at the previous timestep.
r	Cube-spawn perturbation bound used in the uniform offset range $[-r, r]$.
σ	Standard deviation; used as the observation noise level in stress-test conditions.
$\mathcal{N}(0, \sigma^2)$	Gaussian noise distribution with mean 0 and variance σ^2.
n	Number of evaluation episodes in a test condition.

Chapter 1

Introduction

1.1 Problem Statement

Dexterous manipulation policies trained only from human demonstrations often fail when they are executed under contact-rich conditions. A demonstration can show how a hand should approach, close around, and move an object, but it does not necessarily teach the policy how to recover when the object slips, contact occurs slightly late, or the hand enters a state that was not present in the demonstration data. This creates a central problem for demonstration-based manipulation: imitation can initialise useful behaviour, but it may not be sufficient for reliable task completion.

This dissertation addresses whether captured human-motion demonstrations can be used as a real-to-simulation (R2S) starting point for a simulated dexterous arm-hand task, and whether reward-based refinement can overcome the limitations of Behavioural Cloning (BC) alone. The implemented task focuses on a simulated grasp-and-transport-to-target sequence using a virtual arm and five-finger hand. The key issue is not only whether the policy can imitate hand closure, but whether it can sustain object manipulation through lift, transport, and target-region completion.

1.2 Background

Dexterous robotic manipulation requires more than moving an end-effector to a point in space. A multi-finger hand must approach an object, establish contact, lift without losing control, transport the object, and satisfy the task goal while maintaining physical consistency. Each stage is affected by geometry, friction, contact timing, and object motion. A cube-transfer task that appears simple for a human therefore becomes a demanding control problem when executed by a simulated seven-degree-of-freedom arm and a five-finger hand.

Learning from demonstration is useful because it can reduce the search space for difficult manipulation tasks. Instead of beginning from random exploratory actions, a policy can be exposed to recorded examples of task-directed behaviour. However, demonstration-only learning remains vulnerable to covariate shift. Once the learner makes a small execution error, it may enter states that were not represented in the demonstrations and has no learned recovery behaviour.

Reinforcement Learning (RL) provides a mechanism for correcting this weakness because the policy receives feedback from its own interaction with the environment. In contact-rich manipulation, however, RL from random initialisation can be inefficient. This motivates a hybrid learning structure in which BC provides early demonstration assistance and Proximal Policy Optimisation (PPO) then refines the policy using task rewards.

1.3 Motivation

The motivation for this dissertation is to examine what captured human motion actually contributes to a simulated dexterous manipulation pipeline. In prototype-based manipulation projects, imitation and reinforcement are often combined without clearly separating their roles. A visually successful trained arm may look convincing, but that alone does not explain whether the demonstration data solved the task, whether reward refinement solved the task, or whether the learned behaviour generalises beyond the fixed setup.

This dissertation therefore separates the learning pipeline into two conditions. Condition A uses BC from recorded demonstrations as the only learning signal. Condition B uses the same demonstrations as early BC assistance, followed by PPO refinement. By comparing these conditions under the same simulation setup, the study identifies whether demonstration-only learning is sufficient or whether reward-based refinement is required.

The project is also motivated by the need to qualify success carefully. The implemented success metric measures whether the cube enters a target-region radius, not whether the cube is stably released after hand opening. The work therefore treats the prototype as a grasp-and-transport-to-target system rather than a full autonomous pick-place-release system. This boundary is important because it prevents the dissertation from overstating the capability of the trained policy.

1.4 Research Aim, Questions, and Hypotheses

The aim of this dissertation is to investigate whether a virtual arm and hand can be trained within a simulation environment using R2S learning from captured human motion, and to evaluate whether the resulting system can perform the basic manipulation sequence of grasping, lifting, transporting, and reaching a target region within the virtual environment.

The research questions are:

RQ1. Can a virtual arm and hand be trained within a simulation environment with R2S learning using captured human motion to accomplish basic grasping, lifting, transporting, and target-region completion?

RQ2. To what extent can the virtual arm and hand reproduce the task-level structure of the human demonstrations and manipulate objects within the simulator environment?

RQ3. How efficient is the learned virtual arm and hand system concerning its performance and training time, and is the system capable of being used in a real-time or near-real-time manner within the environment?

The research hypothesis is:

> Using R2S learning with captured human-motion data, a virtual arm and hand will be able to successfully execute actions related to grasping, lifting, transporting, and target-region completion within the virtual environment, while operating at a speed sufficient for real-time or near-real-time control.

The null hypothesis is:

Using R2S learning with captured human-motion data, a virtual arm and hand will not achieve successful grasping, lifting, transporting, and target-region completion within the virtual environment, or will not operate at a speed sufficient for real-time or near-real-time control.

The objectives are:

1. Implement a simulated dexterous manipulation training environment with a seven-degree-of-freedom arm, a five-finger hand, a fixed cube spawn, and a target marker.

2. Design an R2S capture pipeline that converts human hand motion into robot-compatible control signals for demonstration recording.

3. Train a BC-only policy (Condition A) using the recorded demonstrations as the sole learning signal.

4. Train a BC-assisted PPO policy (Condition B) using the same demonstrations as early behavioural assistance, followed by reward-based refinement.

5. Compare both policies using target-region success, time-to-success, proxy grasp-validation behaviour, drop behaviour, episode length, and qualitative behavioural analysis.

6. Evaluate the robustness of the trained Condition B policy under controlled inference-time perturbations, including cube-spawn offsets and observation noise.

7. Identify limitations in the implemented reward design, proxy grasp validation, and target-region success criterion.

1.5 Scope and Limitations

The dissertation focuses on a simulated single-arm, five-finger grasp-and-transport-to-target task. The primary A/B comparison uses a fixed cube spawn and a fixed target marker so that the learning regimes can be compared under controlled conditions. Robustness is then evaluated separately by testing the final Condition B policy under controlled inference-time perturbations.

The main success criterion is target-region completion: the cube must enter the placement radius around the marker. Controlled release is not part of the main success metric. Therefore, the work evaluates simulated grasp-and-transport-to-target behaviour rather than full pick-place-release.

The study is also bounded by its implementation context. It does not perform physical robot deployment, tactile sensing, sim-to-real transfer, multi-object manipulation, or multi-seed statistical testing. Detailed methodological boundaries, variables, and exclusions are presented in Chapter 3.

1.6 Dissertation Structure

- **Chapter 2** reviews the literature on dexterous manipulation, Imitation Learning, BC fragility, PPO refinement, and R2S demonstration pipelines.
- **Chapter 3** describes the methodology and implementation, including the research design, observation and action spaces, reward function, experimental

variables, and evaluation framework.

- **Chapter 4** presents and discusses the results, including the A/B comparison, the robustness evaluation under spatial perturbation and observation noise, and the limitations of the success and grasp-validation metrics.

- **Chapter 5** concludes the dissertation, evaluates the contribution, and outlines future work including release-stability validation, broader robustness testing, improved grasp validation, and repeated training seeds.

Chapter 2

Literature Review

2.1 Overview: The Engineering Challenge of Dexterity

Dexterous manipulation requires the coordinated control of a robot arm, a multi-fingered hand, and a moving object under discontinuous contact dynamics [2, 3, 4]. Unlike simple end-effector positioning, a dexterous task requires the system to approach the object, establish stable contact, lift it, transport it, and satisfy a goal condition without losing control of the object. Small contact errors can change the entire state of the task: a fingertip that is only slightly misaligned can push the object away, prevent a lift, or destabilise transport. This sensitivity makes dexterous manipulation difficult for classical hand-designed control, especially when the hand has many degrees of freedom and the relevant contact state is only partially observable [2, 3].

The KUKA iiwa7 and SCHUNK SVH embodiment used in this dissertation reflects this challenge [5, 6]. The arm provides seven degrees of freedom, while the five-finger hand introduces additional contact-rich control requirements. A successful policy must therefore coordinate global reaching with local finger-object interaction. The literature increasingly treats this type of problem as a learning-pipeline problem

rather than a single-controller problem: human demonstrations reduce exploration difficulty, simulation provides safe repeated practice, and reinforcement learning can refine behaviours that demonstrations alone do not make robust [7, 8, 9]. Figure 2.1 illustrates the synthesis route this chapter follows.

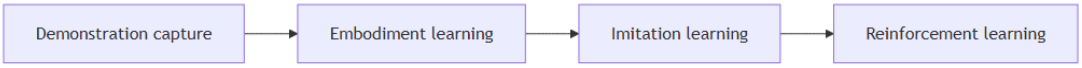


Figure 2.1: Literature synthesis route used to organise the review.

This review focuses on sources directly addressing demonstration capture, embodiment mapping, imitation learning, reinforcement learning refinement, and real-to-simulation training [10, 9, 7, 8]. Peripheral strands such as bimanual manipulation [11, 12], tactile-only approaches [13], language-conditioned manipulation [14], and articulated tool use [15] are recorded in Appendix B because they change the sensing or task assumptions beyond the implemented dissertation scope.

2.1.1 Search Index and Source Matrix

Table 2.1 records the five source clusters used to organise the literature search, listing representative works for each cluster and the reason each cluster was included. The matrix provides a transparent audit trail of the search strategy and ensures that each body of literature can be traced to a specific role in the experimental design.

Table 2.1: Search index and source-cluster matrix for the literature review.

Source cluster	Representative sources	Reason for inclusion
Demonstration capture	DexCap [10]; AnyTeleop [16]; Open Teach [17]	Defines the capture-fidelity baseline and instrumentation trade-off.
Vision-based hand capture	DexMV [9]; One-to-Multiple [18]; Robotic Telekinesis [19]	Supports the use of vision-derived human hand intent as a source of robot supervision.
Minimal-demo and R2S learning	Lum et al. [20]; RialTo [7]; DexTransfer [21]	Supports the argument that small demonstration budgets can seed useful policies when refinement is available.
BC and RL refinement	DIME [8]; RialTo [7]; DexMV [9]	Motivates the BC-only versus BC-assisted PPO experimental comparison.
Peripheral strands	DexMan [11]; ManipTrans [12]; DexTOG [14]; KineDex [13]	Relevant to dexterous manipulation but outside the central single-arm, vision-based, sim-only comparison.

2.2 Machine Learning Paradigms for Robotic Control

Classical robotic control depends on explicit models of robot kinematics, dynamics, and contact mechanics [22]. These approaches are reliable when the state is known and the physics is smooth. Dexterous manipulation violates both assumptions. Contact is intermittent, friction-sensitive, and difficult to observe directly. Model-free learning methods therefore become attractive because they learn mappings from observations to actions without requiring a complete analytical contact model [2, 3, 4].

The standard formalism is the Markov Decision Process (MDP), defined by the tuple $(S, A, T, R, \gamma, \rho_0)$ where:

- S is the state or observation space,
- A is the action space,
- T is the transition function,
- R is the reward function,
- γ is the discount factor,
- ρ_0 is the initial-state distribution [23].

A policy π_θ maps states to actions, and the learning objective is to maximise expected discounted return. The implementation-specific MDP for this dissertation is detailed in Chapter 3.

Imitation learning and reinforcement learning appear repeatedly in the reviewed literature as complementary paradigms for dexterous manipulation [8, 2, 4]. Imitation learning uses expert demonstrations to train a policy directly; reinforcement learning improves a policy through trial-and-error interaction with a reward function. The dissertation's experimental design tests these paradigms as a sequence: BC first imitates captured human motion, then PPO refines the resulting policy.

> **Section synthesis.** The MDP formalism provides the mathematical language for comparing learning approaches, but it does not remove the physical difficulty of contact-rich manipulation. The practical choice is therefore between imitation-based initialisation and reward-based refinement, or a hybrid of both.

2.3 Skill Acquisition via Imitation Learning

Imitation Learning allows a robot to acquire task structure from expert demonstrations instead of discovering the whole behaviour through random exploration [24, 2]. Behavioural Cloning (BC) is the simplest form of this approach. Given demonstration pairs $\mathcal{D} = \{(s_i, a_i^*)\}$, BC trains a policy by minimising the difference between the demonstrated action and the predicted action:

$$\mathcal{L}_{\mathrm{BC}}(\theta) = \mathbb{E}_{(s,a^*)\sim\mathcal{D}} \left[\|\pi_\theta(s) - a^*\|^2 \right]. \tag{2.1}$$

BC is attractive because it is data-efficient and does not require reward design. It can move the policy into a behaviourally meaningful region of the action space before RL begins [20, 7, 21]. Its weakness is that it only learns from states represented in the demonstrations. The policy therefore lacks an explicit mechanism for recovering from errors that move it outside the demonstrated trajectory distribution [25]. Figure 2.2 illustrates how BC and PPO address complementary failure modes within this pipeline.

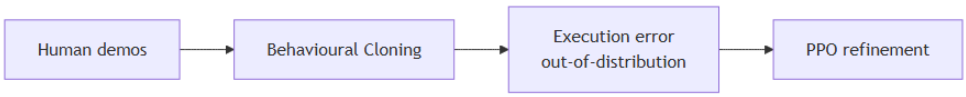


Figure 2.2: BC and RL address different failure modes. BC supplies the behavioural prior; PPO refinement learns corrective behaviour through environmental interaction.

Section synthesis. Behavioural Cloning provides a direct and data-efficient way to transfer task intent from human demonstrations. However, it only learns from states that appear in the demonstration dataset. When the policy encounters its own execution errors and drifts outside that distribution, it has no built-in recovery mechanism – a weakness that motivates the analysis of BC fragility in the next section.

2.4 The Problem of Information Fragility

BC is vulnerable to covariate shift [25, 2]. During training, the policy sees expert states; during deployment, it sees states produced by its own imperfect actions. If a small error changes the contact state of the cube, the policy may enter a region with no demonstrated recovery action. In dexterous manipulation this is severe because the success region for a stable grasp is narrow [8, 7].

The empirical literature reflects this fragility. Arunachalam et al. [8] report 0% task success for a vanilla BC baseline across four dexterous manipulation tasks, while their improved method recovers success up to 75%. Torne et al. [7] report BC-only success of 5–25% on real-world tasks, rising to 75–91% after RL refinement. DexTransfer [21] and DexMV [9] show the same directional pattern: demonstration-only performance is weakest where contact execution is most sensitive, and refinement produces the largest gains.

A small demonstration set may be enough to show the intended sequence of reaching, grasping, lifting, and transporting, but it is unlikely to cover all contact errors or recovery states. This follows the covariate-shift problem identified by Ross et al. [25]: during deployment, an imitation policy encounters states induced by its own imperfect actions rather than only the expert states seen during training. In dexterous manipulation, this problem is amplified because small contact errors can change the object state and require recovery behaviours that were not present in the demonstrations [8, 7].

Section synthesis. BC provides efficient task initialisation, but it is incomplete as a control solution for contact-rich manipulation. Its failure mode is not lack of task intent; it is lack of recovery outside the demonstration manifold.

The foregoing establishes why imitation alone is insufficient for the present task: it cannot generate the recovery signals needed to handle contact errors that fall outside the demonstration distribution. A suitable complement is a learning mechanism that generates experience through interaction with the environment and receives reward for correcting or avoiding those errors. Reinforcement learning provides this mechanism, and stochastic policy-gradient methods are particularly relevant because they directly optimise a parameterised policy from rollout experience.

2.5 Stochastic Policy Refinement via Reinforcement Learning

Reinforcement Learning optimises expected return under the policy's own execution. This matters because the agent receives feedback from states that were not included in the demonstrations. For dexterous manipulation, RL from random initialisation is usually inefficient, so the literature commonly uses BC to initialise the policy and RL to refine it [8, 7, 9, 3].

PPO is used in this dissertation because it provides conservative policy updates through a clipped surrogate objective [26]. Given the probability ratio

$$r_t(\theta) = \frac{\pi_\theta(a_t \mid s_t)}{\pi_{\theta_{\text{old}}}(a_t \mid s_t)}$$

and advantage estimate $\hat{A}_t$, PPO optimises the clipped surrogate objective:

$$\mathcal{L}^{\text{CLIP}}(\theta) = \mathbb{E}_t \left[\min \left(r_t(\theta)\hat{A}_t, \text{clip}(r_t(\theta), 1-\epsilon, 1+\epsilon)\hat{A}_t \right) \right]. \tag{2.2}$$

The expectation is over sampled timesteps from policy rollouts. The clipping parameter, set to $\epsilon = 0.2$ in this study, limits destructive jumps away from the current policy.

This is useful when refining a BC-seeded policy because the useful demonstration prior should be improved, not destroyed.

Table 2.2: Representative BC-only versus hybrid BC→RL results from reviewed studies.

Study	Capture method	BC-only result	After RL refinement
DIME [8]	Teleoperation	0% on all tasks	Up to 75% success
RialTo [7]	Real demonstrations	5–25% success	75–91% success
DexTransfer [21]	2–3 MoCap demos	11% success	61–74% success
DexMV [9]	Human task video	0.11–0.67 score	0.98 on hardest reported task

Section synthesis. The BC→RL pattern is the strongest literature-supported response to BC fragility. BC supplies task structure; PPO supplies corrective adaptation through reward feedback.

2.5.1 The Hybrid BC→RL Pattern

Across DIME, RialTo, DexTransfer, and DexMV, the same broad pattern appears: hybrid BC→RL pipelines outperform BC-only baselines in contact-rich dexterous manipulation, although the size of the gain depends on demonstration quality, task difficulty, and the refinement reward design. Arunachalam et al. [8] report that vanilla BC achieved 0% success across four dexterous tasks, whereas the improved hybrid method reached up to 75%. Torne et al. [7] report a similar improvement pattern, with BC-only performance of 5–25% increasing to 75–91% after refinement on real-

world tasks. DexTransfer [21] and DexMV [9] show the same directional pattern: demonstration-only performance is weakest where contact execution is most sensitive, while refinement produces the largest gains.

The main explanation is covariate shift: the policy is trained on states from the demonstration dataset, but during execution it encounters states produced by its own imperfect actions. Once the policy drifts into unseen states, it has no explicit recovery mechanism [25]. RL refinement addresses this weakness by exposing the policy to its own execution distribution and providing corrective feedback through the reward signal. Second, demonstration data can contain irrelevant variation or capture noise, especially when collected using lower-cost sensing pipelines. BC directly fits the demonstrated action distribution, while RL can optimise for task outcomes such as grasp, lift, transport, and target-region completion rather than exact low-level trajectory reproduction [7, 9].

The hybrid pattern is not unconditional. Demonstration quality affects robot learning and generalisation [27]; if demonstrations are incomplete, inconsistent, or too far from the desired task manifold, BC initialisation may not place the policy in a useful region for later RL refinement. Conversely, if the task is low-dimensional or highly constrained, imitation alone may already provide a usable controller, reducing the value added by RL. The present dissertation occupies a middle ground: the task is constrained to a fixed grasp-and-transport-to-target layout, but the embodiment still involves a five-finger hand-arm system and contact-rich object interaction. The literature therefore predicts that BC alone may be fragile, while BC-assisted PPO should produce a measurable refinement gain. This hypothesis is tested experimentally in Chapter 3.

2.6 The Real-to-Simulation Paradigm

Real-to-Simulation (R2S) pipelines capture human motion in the real world and convert it into simulation-compatible supervision [7, 2, 28, 29]. This reduces the need to explore physically on hardware while still grounding the policy in human task intent [7, 21]. Capture modality is a major design decision because it determines the fidelity, cost, and accessibility of the demonstration signal [10, 16, 17, 9]. Table 2.3 summarises the reviewed methods on this cost–fidelity axis.

Table 2.3: Capture-method trade-off across the reviewed literature. The dissertation uses an intermediate Kinect–MediaPipe position: more instrumented than a basic webcam, less costly than MoCap or dedicated teleoperation.

Capture method	**Fidelity**	**Cost**	**Used in study?**
RGB webcam (MediaPipe-class)	Low	Very low	No
Kinect + MediaPipe	Medium	Low	**Yes**
Teleoperation rig (AnyTeleop, Open Teach)	Medium–high	Medium	No
MoCap gloves (DexCap-class)	High	High	No

2.6.1 Capture Fidelity, Demonstration Quality, and Policy Initialisation

The reviewed literature also shows that demonstration quality should not be understood only as raw sensor accuracy. A high-fidelity capture system can record detailed motion, but the resulting data must still be transformed into a representation that the robot can execute [9, 16]. Conversely, a lower-cost capture system may lose fine hand-pose detail but remain useful if it preserves the task-relevant sequence of ap-

proaching, closing, lifting, and transporting [9]. This distinction matters because the present dissertation does not attempt full anatomical reconstruction of the human hand. It uses captured hand intent as a practical supervisory signal for a constrained simulated task.

DexCap, AnyTeleop, and Open Teach represent stronger capture and teleoperation baselines because they provide cleaner kinematic supervision and more controlled embodiment mappings [10, 16, 17]. However, these systems also require more hardware, calibration, and experimental setup than the resource-constrained workflow used in this dissertation. Vision-based approaches such as DexMV [9] and Robotic Telekinesis [19] show that human task intent can be extracted from video, but the recovered signal is less precise and more vulnerable to occlusion or jitter [30, 31].

A vision-based hand-tracking interface can localise hand landmarks from RGB input and convert the resulting signal into a lower-dimensional control representation [32, 33]. Such an interface does not aim for motion-capture-level fidelity. Instead, its value lies in whether it preserves sufficient task intent to initialise a behavioural prior that PPO can later refine.

2.6.2 Kinematic Abstraction and Embodiment Gap

Human hands and robot hands differ in morphology, joint coupling, and workspace [34, 16, 10]. Demonstration capture therefore requires retargeting. The reviewed systems address this with different abstractions: some use detailed MoCap retargeting, while others compress hand motion into task-relevant features [10, 16, 9]. Retargeting from captured hand landmarks to robot finger commands is a common strategy in R2S pipelines [10, 9]. This dissertation adopts a similar abstraction, implemented via

ML-Agents, with the implementation detail deferred to Chapter 3.

Section synthesis. Real-to-simulation pipelines enable training from captured human motion without hardware risk. Capture fidelity and retargeting are key design choices: high-fidelity systems provide cleaner supervision but require specialised hardware, while lower-cost vision-based capture (Kinect–MediaPipe) loses fine detail but may still supply a useful behavioural prior. This trade-off is central to the dissertation's methodology, which tests whether such a low-cost pipeline can initialise a policy that PPO can later refine into successful target-region transport.

2.7 Synthesis of Reviewed Literature

The reviewed work spans four interconnected themes: learning paradigms, policy fragility, refinement mechanisms, and real-to-simulation capture. The first theme concerns the modelling of robotic control as an MDP and the use of imitation learning and reinforcement learning within that framework. The second concerns the fragility of BC under covariate shift, where policies trained only on expert states fail when their own execution errors move them outside the demonstrated state distribution. The third concerns PPO and hybrid BC→RL refinement, where demonstrations provide an initial behavioural prior and reward-based interaction supplies corrective adaptation. The fourth concerns R2S capture pipelines, where human motion is converted into simulation-compatible supervision but must pass through capture-fidelity, retargeting, and embodiment-mapping constraints.

Across these themes, two recurring patterns emerge. First, BC-only baselines often fail or underperform on dexterous manipulation because demonstration data transfers task intent but does not provide robust recovery behaviour. Second, hybrid BC→RL approaches can recover from BC-only fragility when the demonstrations place the policy near a useful region of the state-action space [8, 7, 21, 9, 20]. This

pattern is especially relevant to the present dissertation because the implemented task is constrained to a fixed grasp-and-transport-to-target layout (target-region completion), but still involves a five-finger hand-arm embodiment and contact-rich object interaction.

The R2S literature further shows that capture quality should not be judged only by sensor fidelity. High-fidelity systems can provide cleaner motion supervision, but they require more specialised hardware and calibration. Lower-cost vision-based capture may lose fine hand-pose detail, but can still be useful if the captured signal preserves task-level intent and is treated as a behavioural prior rather than a complete controller. This frames the central methodological question of the dissertation: whether a low-cost Kinect–MediaPipe capture pipeline can initialise a policy well enough for PPO to refine it into successful target-region transport behaviour.

2.7.1 Research Gap

The gap addressed by this dissertation is the combination of constraints: low-cost vision-based capture, a five-finger hand-arm embodiment, a small demonstration set, and a controlled sim-only BC-only versus BC-assisted PPO comparison. Within the reviewed literature, no study was identified that reports a controlled, sim-only comparison of BC-only versus BC-assisted PPO for a five-finger hand-arm system using a low-cost, vision-based real-time hand capture pipeline. The closest works (RialTo [7], DexMV [9], DexCap [10], AnyTeleop [16], DexTransfer [21], Lum et al. [20]) each differ on at least two of: capture instrumentation, embodiment, or evaluation scope.

Beyond the baseline comparison, the reviewed studies generally emphasise task success under their main evaluation conditions, while offering limited direct quantifi-

cation of robustness to spatial perturbations or observation noise. To address this, the present dissertation includes a stress-test evaluation (Section 4.7) that measures policy success under controlled offsets of the cube spawn, additive Gaussian observation noise, and combined perturbations.

Figure 2.3 maps this intersection.

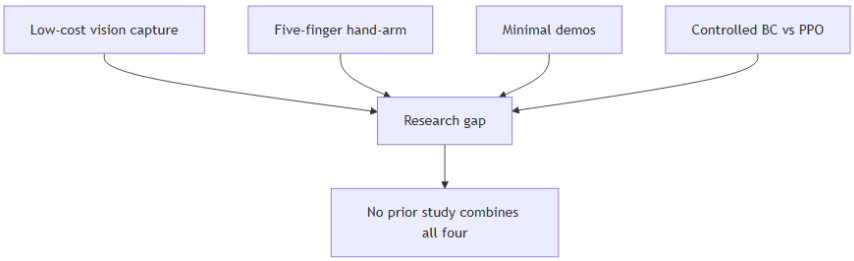


Figure 2.3: Research gap identified from the reviewed literature.

This dissertation contributes a controlled descriptive comparison of BC-only versus BC-assisted PPO under a low-cost R2S pipeline, reporting the observed learning-regime gain for a five-finger hand-arm system under identical simulation conditions.

2.8 Literature Synthesis and Methodological Positioning

The reviewed literature positions dexterous manipulation as a pipeline problem rather than a single-algorithm solution. Demonstrations provide task intent, Behavioural Cloning turns that intent into an initial policy, Proximal Policy Optimisation supplies corrective refinement, and simulation offers a safe environment for repeated interaction [35, 7]. Two recurring patterns emerge: BC-only baselines consistently underperform on contact-rich tasks due to covariate shift, while hybrid BC→RL pipelines recover performance even with small demonstration budgets [8, 7, 21, 9].

However, the combination of constraints central to this dissertation has not

been directly examined in the reviewed literature: low-cost, vision-based hand capture (MediaPipe + Kinect), a five-finger hand-arm embodiment (KUKA iiwa7 + SCHUNK SVH), a minimal demonstration budget, and a strictly sim-only controlled comparison. Closely related works (RialTo, DexMV, DexTransfer, Lum et al.) differ on at least two of these dimensions, leaving the specific performance gap unquantified.

The following chapter therefore operationalises the two experimental conditions, BC-only versus BC-assisted PPO, within the Unity ML-Agents framework, keeping all other variables fixed. Chapter 3 details the MDP formalisation, reward structure, demonstration capture pipeline, and evaluation metrics. In addition to the baseline A/B comparison, the evaluation includes a stress-test framework (Chapter 4) that assesses policy robustness under cube-spawn perturbations up to $\pm 15\,\text{cm}$, additive observation noise, and a combined $\pm 6\,\text{cm}$ spatial-offset plus low-noise condition. This directly addresses the absence of systematic robustness assessment identified in the literature and provides a more complete empirical picture than a single-condition success rate alone.

Chapter 3

Research Methodology

3.1 Overview

This chapter explains the research design and implementation used to evaluate a demonstration-based real-to-simulation (R2S) pipeline for dexterous manipulation. In this dissertation, R2S refers to transferring captured real human hand-motion intent into a simulated robotic manipulation environment, rather than transferring a physical robot policy into simulation. The prototype uses a simulated KUKA iiwa7 seven-degree-of-freedom arm and a SCHUNK SVH five-finger hand in Unity 2022.3 LTS. Human hand intent is captured with a Kinect-based vision input pipeline and converted into low-dimensional hand-control signals that are consumed by Unity during demonstration recording. The trained agent then controls the simulated robot through Unity ML-Agents using a continuous action space.

The empirical comparison consists of two conditions: Condition A: BC-only and Condition B: BC-assisted PPO refinement. Both conditions use the same task scene, robot embodiment, demonstration dataset, observation layout, action layout, and target-region success criterion. The main focus is whether reward-based PPO refinement improves a policy trained with early demonstration assistance from cap-

tured human motion. The evaluation boundary and interpretation of the completed A/B comparison are defined in the descriptive analysis protocol (Section 3.7.3).

3.1.1 Research Aim

The aim of the study is to investigate whether a virtual arm and hand can be trained within a simulation environment using real-to-simulation learning from captured human motion, and to evaluate whether the resulting system can perform the basic manipulation sequence of grasping, lifting, transporting, and target-region completion within the virtual environment.

3.1.2 Research Questions and Hypotheses

The study is guided by three research questions:

RQ1. Can a virtual arm and hand be trained within a simulation environment with real-to-simulation learning using captured human motion to accomplish basic grasping, lifting, transporting, and target-region completion?

RQ2. To what extent can the virtual arm and hand manipulate objects within the simulator environment, and does the learned policy reproduce the task-level structure of the human demonstrations?

RQ3. How efficient is the learned virtual arm and hand system concerning its performance and training time, and is the system capable of being used in a real-time or near-real-time manner within the environment?

The research hypothesis is:

Using real-to-simulation learning with captured human-motion data, a virtual arm and hand will be able to successfully execute actions related

to grasping, lifting, transporting, and target-region completion within the virtual environment, and operate at a speed sufficient for real-time or near-real-time control.

The null hypothesis is:

Using real-to-simulation learning with captured human-motion data, a virtual arm and hand will not achieve successful grasping, lifting, transporting, and target-region completion within the virtual environment, or will not operate at a speed sufficient for real-time or near-real-time control.

Within the implemented A/B testing matrix, these hypotheses are operationalised through the comparison between a BC-only baseline and a BC-assisted PPO policy. Condition A tests whether demonstration matching alone is sufficient to produce target-region transport behaviour, while Condition B tests whether BC assistance followed by PPO refinement improves the same demonstration prior.

3.1.3 Research Scope

The empirical scope is a vision-based R2S workflow for a single-arm, five-finger dexterous manipulation task in simulation. The primary A/B training comparison uses a fixed cube spawn and a fixed target marker so that the comparison isolates the completed learning pipeline rather than task randomisation. Spawn variation is introduced separately during the inference-time robustness evaluation, where the trained Condition B policy is tested under controlled cube-spawn perturbations without retraining. The implemented success criterion is target-region completion: the cube must reach the target-region radius (set to 0.15 m). Controlled release is not enforced

in the main result; therefore, the system is evaluated as grasp-and-transport-to-target behaviour rather than solved full pick-place-release behaviour.

The study does not perform physical robot deployment, sim-to-real transfer, tactile sensing, bimanual manipulation, visual distractor robustness testing, or a multi-task manipulation benchmark. These exclusions keep the experiment focused on the BC versus BC-assisted PPO comparison.

3.1.4 Research Limitations

The following boundaries define what the completed empirical work does not address:

- Physical robot deployment and sim-to-real transfer.
- Controlled object release (target-region success is used as the primary criterion instead).
- Tactile sensing, bimanual manipulation, object classification, throwing, bouncing, or stacking.
- Broad randomised multi-scene generalisation across different target positions, scene layouts, object types, or distractor-object configurations; limited cube-spawn perturbations are included only as an inference-time robustness stress test.

These boundaries keep the methodology focused on the implemented A/B comparison and the controlled robustness evaluation.

3.2 Research Framework and Formalisation

3.2.1 Philosophical Positioning

This study adopts a post-positivist, deductive, quantitative research design. The hypothesis is directional: BC-assisted PPO is expected to outperform BC-only learning under the same fixed task scene and success criterion. The experiment controls the task scene, robot embodiment, demonstration dataset, observation space, action space, and target-region success criterion, while varying the completed learning pipeline. Performance is evaluated using numerical task metrics, including target-region success rate, time-to-success, proxy grasp-validation rate, lift rate, drop rate, timeout rate, and minimum distance to target.

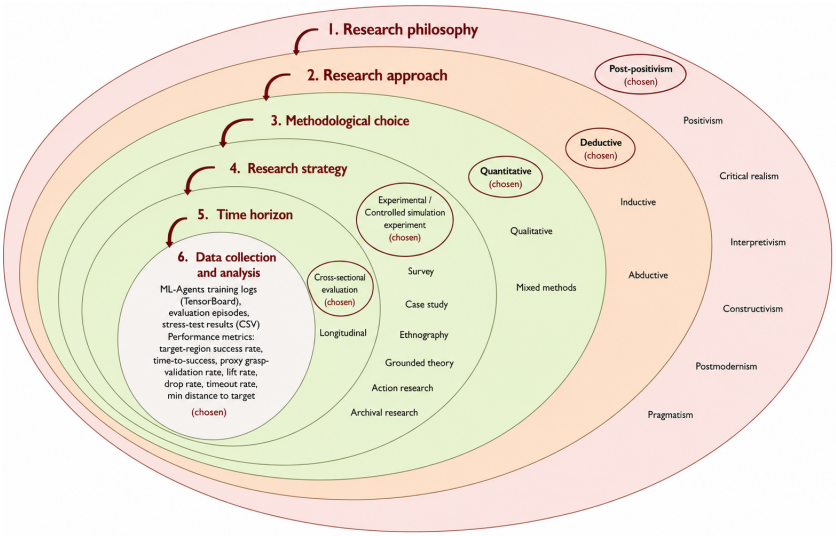


Figure 3.1: Research onion framing for the dissertation methodology, adapted from Saunders et al.'s layered research-design model [1]. Each concentric layer represents a methodological commitment, from the broadest philosophical stance (outer) to the data collection and analysis approach (core).

Figure 3.1 summarises the methodological positioning of this dissertation. The outer layer identifies the research philosophy as post-positivist because the study uses objective performance metrics while acknowledging simulation boundaries, proxy val-

idation limits, and single-run statistical constraints. The research approach is deductive because the experiment tests the literature-derived hypothesis that BC-assisted PPO should outperform BC-only learning under the same fixed task definition. The methodological choice is quantitative because the evaluation relies on numerical task metrics. The research strategy is experimental, using a controlled simulation environment in which the completed learning pipeline is varied while the robot embodiment, task scene, observation space, action space, and demonstration dataset remain fixed. The time horizon is cross-sectional because the final trained policies are evaluated over fixed episode sets, while TensorBoard logs are used to interpret the training process. Data collection and analysis consist of ML-Agents training logs, TensorBoard reward and episode trends, evaluation CSV files, task success rate, time-to-success, proxy grasp-validation rate, physical lift rate, drop rate, timeout rate, minimum distance to target, and the eight-condition robustness stress-test results.

3.2.2 Experimental Design

The research design is quantitative and experimental. It compares two completed learning pipelines under the same fixed task scene, robot embodiment, observation space, action space, demonstration dataset, and target-region success criterion. The design is deductive because it tests the literature-derived expectation that BC-only policies are fragile under contact-rich manipulation and that PPO refinement can improve the demonstration prior.

3.2.3 Experimental Variables

The main independent variable is the completed learning pipeline: BC-only learning in Condition A versus BC-assisted PPO refinement in Condition B. Because Condi-

tion B includes an extended PPO refinement phase, the comparison is not an equal-budget ablation. Condition A used ML-Agents Behavioural Cloning from the recorded demonstrations only; no extrinsic task reward contributed to the Condition A learning signal. Condition B uses the same demonstration set as a BC-assisted prior within a PPO trainer (auxiliary BC loss active in early training, decaying to zero) followed by PPO-only refinement.

The dependent variables are task success rate, mean time-to-success, proxy grasp-validation rate, physical lift rate, drop rate, timeout rate, and minimum distance to the target marker. A validated grasp is treated as an internal proxy event, not as proof that the cube was physically secured; the evaluation framework separates the grasp flag from physical lift and target-region success to prevent equating them.

For the primary A/B training comparison, the controlled variables include the robot model, Unity scene, fixed cube spawn, fixed target marker, physics timestep, observation and action spaces, network architecture, demonstration dataset, and environment reset procedure. Trainer-specific settings are reported separately. Cube-spawn variation is introduced only in the separate inference-time robustness evaluation, where the converged Condition B policy is stress-tested without retraining. The final implementation uses a 27-dimensional observation vector and a 13-dimensional continuous action vector.

Table 3.1: Experimental variable structure used in the final implementation.

Variable class	Operational definition
Independent variable	Completed learning pipeline: Condition A (BC-only) versus Condition B (BC-assisted PPO refinement).
Dependent variables	Success rate, time-to-success, proxy grasp-validation rate, physical lift rate, drop rate, timeout rate, and minimum distance to target.
Controlled variables for A/B comparison	Robot model, fixed task layout, fixed cube spawn, target marker, demonstration set, observation/action spaces, network architecture, physics timestep, and environment reset procedure. Trainer-specific settings are reported separately; cube-spawn perturbation is introduced only during the robustness stress test.
Evaluation boundary	Target-region success is measured; controlled object release is not enforced in the main comparison.

3.2.4 Problem Formalisation as a Discrete-Time MDP

The task is formalised as a discrete-time Markov Decision Process (MDP) defined by the tuple $(S, A, T, R, \gamma, \rho_0)$. S is the continuous observation space, A is the continuous action space, T is the Unity physics transition function, R is the task reward, γ is the discount factor, and ρ_0 is the initial state distribution. The transition model is not learned explicitly; it is supplied by the Unity physics engine through environment interaction.

In the final implementation, $S \subset \mathbb{R}^{27}$. The 27 observations consist of seven normalised arm articulation target values, six hand-state values, the cube position relative to the palm, the target position relative to the cube, a validated-grasp flag, a hold/attachment flag, cube linear velocity, and cube angular velocity. The arm

values are the current desired joint targets used by the ArticulationBody drives, normalised to $[-1, 1]$ by dividing degrees by 180; they are not raw torque readings. The action space is $A \subset [-1, 1]^{13}$, with seven arm joint-delta actions and six hand actions controlling thumb, index, middle, ring, pinky, and spread.

The BC objective is supervised action matching over the demonstration dataset:

$$\mathcal{L}_{\mathrm{BC}}(\theta) = \mathbb{E}_{(s,a^*)\sim\mathcal{D}} \left[\|\pi_\theta(s) - a^*\|^2 \right]. \tag{3.1}$$

The PPO objective instead maximises expected return under the reward function, using a clipped surrogate objective to limit destructive updates during policy refinement.

3.3 Implementation

3.3.1 Architecture

The prototype is built as an integrated R2S pipeline across three phases: demonstration, training, and evaluation. The demonstration dataset consisted of 60 recorded demonstration episodes. During demonstration, a human operator performs the intended hand motion while the capture pipeline converts the motion into hand-control values. Unity records the resulting observations and actions using the ML-Agents demonstration system. During training, the same demonstration set is used either as the only learning signal (Condition A) or as a BC-assisted prior for PPO refinement (Condition B). During evaluation, trained checkpoints are analysed using TensorBoard training logs and, for the final Condition B policy, inference-mode stress-test episodes scored using the same task metrics.



Figure 3.2: Implemented R2S pipeline: human capture feeds the Unity R2S interface, which records ML-Agents demonstration files used for training under Condition A (BC-only) and Condition B (BC-assisted PPO refinement); the trained checkpoints are then analysed using task-level metrics from training logs and, for Condition B, inference-mode robustness tests.

3.3.2 Vision-Based Capture

The demonstration interface uses a Kinect sensor as the visual capture device. The captured hand landmarks are converted into hand-control values and transmitted to Unity as a seven-float UDP packet: thumb, index, middle, ring, pinky, spread, and wrist roll. Wrist roll is received and logged for debugging but is not used to drive the robot arm in the final training setup. The active hand controls are therefore the five finger flexion values plus spread. Per-finger flexion is computed as the sum of interior bend angles at the MCP, PIP, and DIP joints, derived from the 21 three-dimensional landmark positions detected by MediaPipe; the total bend is normalised to $[0, 1]$ using empirical caps observed during calibration in this implementation (approximately 195° for fingers and 85° for the thumb). These caps are used as signal-normalisation limits rather than anatomical joint-range claims. Spread is derived from the angular separation between the index-MCP and pinky-MCP vectors from the wrist landmark, normalised to $[0, 1]$ at 45°.

The use of a Kinect-based vision input keeps the system lower-cost and less instrumented than MoCap gloves or force-feedback teleoperation rigs. The trade-off is reduced capture fidelity compared with dedicated robotic teleoperation systems. For this reason, the signal is abstracted into robot-compatible scalar controls rather than attempting full anatomical retargeting of every human hand joint.

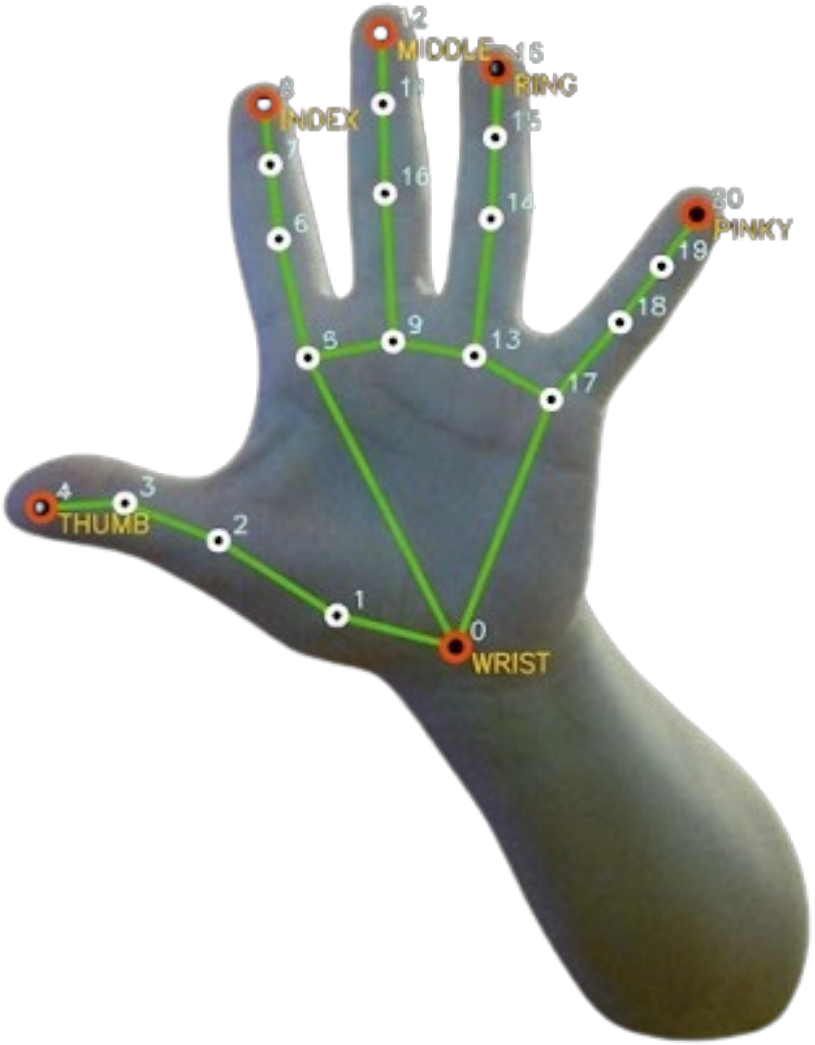


Figure 3.3: MediaPipe hand tracking overlay showing 21 landmarks on the operator's hand. The scalar flexion values for thumb, index, middle, ring, pinky, and spread are derived from landmark-based bend angles and angular separation measures before being transmitted to Unity.

3.3.3 Kinematic Abstraction

The system bridges the embodiment gap by converting human hand intent into a compact set of robot-compatible control values. Instead of attempting to copy all human joint angles, the final implementation uses per-finger scalar control values for thumb, index, middle, ring, pinky, and spread. These values drive the SCHUNK SVH hand in Unity. This abstraction provides separate per-finger control while avoiding

the complexity of full human-to-robot anatomical retargeting.

The KUKA iiwa7 arm provides the transport component of the task, while the methodological focus remains the learning regime and the hand-object interaction.

3.3.4 Unity ML-Agents Training Environment

The environment is implemented in Unity 2022.3 LTS using Unity ML-Agents. The robot model consists of a KUKA iiwa7 arm and SCHUNK SVH hand imported and configured with Unity articulation bodies. The task scene contains a cube at a fixed spawn position and a fixed target marker on the destination table. The physical lift threshold is set to 0.1 m above the table surface, and the episode timeout is 500 decision steps under the configured ML-Agents decision interval. A drop event is recorded when the cube contacts the floor boundary or, following a validated grasp, returns to within 0.15 m of the spawn position below the lift threshold. Each episode begins from the same task layout to ensure that the comparison between conditions is not confounded by environment randomisation.

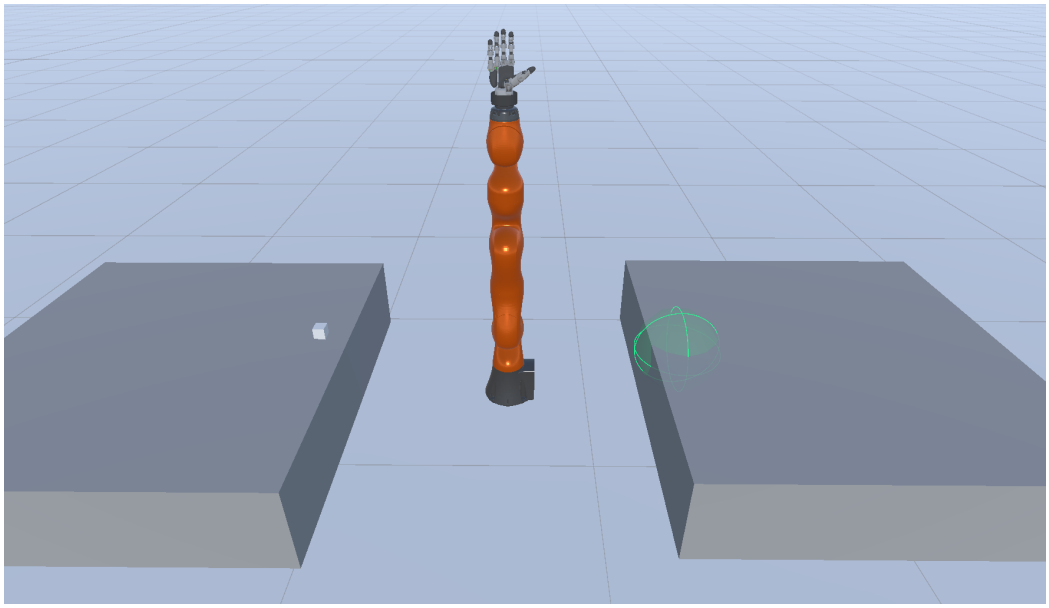


Figure 3.4: Unity simulation environment. The KUKA iiwa7 arm and SCHUNK SVH hand are shown with the cube at the fixed spawn location (left) and the target marker (right, green radius). The hand is in an open pose before demonstration recording.

Control Interface

The control interface is the boundary between the trained policy and the simulated robot. At each Unity decision step, the policy receives the 27-dimensional observation vector and outputs a 13-dimensional continuous action vector in the range $[-1, 1]$. The arm actions are applied as joint-target deltas to the KUKA iiwa7 ArticulationBody drives; the hand actions are applied as direct control values to the SCHUNK SVH fingers and spread. This design decouples the policy output from raw torque or position commands, which simplifies the action space and improves training stability.

Table 3.2: Final observation-space composition.

Observation group	Dimensions	Purpose
Arm targets	7	Normalised current ArticulationBody target values for the iiwa7 joints.
Hand state	6	Thumb, index, middle, ring, pinky, and spread values.
Palm-to-cube vector	3	Egocentric object location relative to the palm.
Cube-to-target vector	3	Direction and distance from cube to target marker.
Task flags	2	Validated-grasp flag and hold/attachment flag.
Cube linear velocity	3	Motion state of the object.
Cube angular velocity	3	Rotational motion state of the object.
Total	**27**	Final implemented observation vector.

Table 3.3: Final continuous action-space composition.

Action group	Dimensions	Purpose
Arm joint deltas	7	Continuous delta commands for the iiwa7 joint targets.
Hand controls	6	Thumb, index, middle, ring, pinky, and spread control.
Total	**13**	Final implemented per-finger action vector.

3.3.5 Grasp Validation Logic

The GraspValidator subsystem provides an internal event signal used for reward shaping and diagnostic metrics. In the final implementation, validation is intentionally simple because ArticulationBody trigger events were unreliable during testing. A grasp is validated when the average hand-close value reaches at least 0.80 and remains stable for at least 0.12 seconds. Note that the GraspValidator component's own `minHandClose` field is initialised to 0.45 in the prefab, but the `DexterousPickPlaceAgent` overrides it to `graspThresholdNormal = 0.80` at the start of every episode in `OnEpisodeBegin()`; the effective runtime threshold for all Condition A and Condition B episodes is therefore 0.80. The script also contains Inspector fields such as `maxPalmDistance = 0.16m` and `maxCubeSpeed = 0.08m/s`, but these checks were not active in the final code path. Therefore, the validator should be interpreted as a closure-stability proxy, not as a full physical contact proof.

This distinction is important because a false positive grasp flag can occur if the hand closes near the cube but fails to lift or transport it. For that reason, the evaluation framework separates proxy grasp-validation rate from physical lift rate, drop rate, and final target-region success rate. Physical lift and target-region success are logged separately to avoid equating a grasp flag with actual task completion.

3.3.6 Reward Function and Reward Scaling

The reward function is the core mechanism through which PPO learns beyond the demonstrations. Condition A disables extrinsic reward and therefore uses the demonstrations as the only learning signal. Condition B enables the reward function so that the agent can refine the demonstration-assisted policy through interaction.

Table 3.4: Main reward components used during PPO refinement (Condition B). Values are as implemented in `DexterousPickPlaceAgent.cs`.

Component	Value	Direction	Purpose
Validated grasp	+0.5 (one-time)	Positive	Rewards the first validated grasp event.
Physical lift	+1.0 (one-time)	Positive	Rewards lifting the cube above the height threshold (0.1 m).
Transport progress	$+0.005 \times (d_{t-1} - d_t)$ per step	Positive/negative shaping	Rewards reduction in cube-to-target distance after lift; negative values penalise movement away from the target.
Target-region success	+3.0 terminal	Positive terminal	Ends the episode when the cube enters the target-region radius (0.15 m).
Drop penalty	−1.5 terminal	Negative terminal	Penalises dropping the cube before successful target-region completion.
Time penalty	−0.0005 per step	Negative shaping	Discourages slow or idle behaviour.

Absolute reward values should be interpreted as the sum of several event and shaping components rather than as direct percentages. A high-return episode can combine multiple positive events, such as validated grasp, lift, transport progress, and target-region success. This explains why isolated reward spikes can be much larger than the near-zero BC baseline. The main performance metric is therefore success rate, supported by reward and time-to-success trends.

The success criterion used in the main comparison is target-region success:

the cube reaches the target-region radius around the marker. The main training result therefore measures grasp-and-transport-to-target rather than controlled release. A stricter release-oriented variant was explored separately, but it produced reward-hacking behaviour in which the agent learned to throw or launch the cube toward the target. For this reason, the release-oriented variant was excluded from the main evaluation, and the final reward design retained target-region completion as the primary success criterion.

3.3.7 Software Testing

Several implementation issues had to be resolved before training produced meaningful behaviour. The most important was collider and contact stability for the SCHUNK SVH hand and the cube. Early versions used collider setups that caused finger instability, contact tunnelling, or unreliable cube interaction. The final workflow uses baked physical colliders for the SVH links and deterministic cube reset logic, with physics state reset at the start of each episode. A second issue was consistency between demonstration recording and training action spaces: the project moved from an older 10-action synergy control setup to a 13-action per-finger control setup. The final methodology reflects the 13-action implementation only.

3.4 Phase I: Pre-Testing and System Verification

3.4.1 Functional Implementation Testing

Pre-testing verified that the robot model imported correctly into Unity, that all articulation bodies were assigned valid joint limits, and that the hand could open and close through its full range without destabilising the scene. The cube spawn and

reset behaviour were also verified: the object returned to its expected position and orientation at the start of each episode without clipping through the floor or collider surfaces.

Keyboard and heuristic control were used before any ML training began. This allowed manual confirmation that the task could be completed end-to-end: reaching, closing the fingers around the cube, lifting above the lift threshold, and transporting toward the target marker. It also confirmed that no scene-level physics failure would prevent successful episodes.

3.4.2 Communication and Latency Check

The capture pipeline was checked by monitoring incoming UDP packets and logging the per-channel values for thumb flexion, individual finger flexions, spread, and wrist roll. Each channel was verified to produce smooth, continuous values within the expected normalised range before demonstration recording began. The wrist-roll channel was retained in the packet structure as a diagnostic value but was not used for final arm control.

Training was not started until the hand-control signal was stable: no rapid jitter, no dropped packets, and no saturated values that could produce inconsistent demonstration actions. A brief soak test was used to confirm pipeline stability before the formal demonstration recording session; this involved operating the capture system for several minutes and inspecting the logged values.

3.4.3 Heuristic Verification

Before recording the final demonstration dataset, the demonstration recording path was verified to ensure that all arm and hand actions were written into the ML-Agents

action buffer rather than applied directly to the robot outside the agent loop. This distinction is critical for behavioural cloning: BC can only replicate actions that are recorded in the demonstration files, so any action that bypasses the agent buffer would be invisible to the learner.

Heuristic tests confirmed that the agent correctly received and buffered both the arm delta actions and the per-finger hand-control actions during the demonstration recording mode. A sample of recorded demonstration files was inspected to verify that the action dimensions matched the expected 13-dimensional continuous action space before the full 60-episode dataset was collected.

3.5 Phase II: Pilot Testing and Model Optimisation

3.5.1 Initial Learning Performance

Pilot training runs were used to confirm whether the two regimes produced non-random behaviour. Condition A was expected to learn the demonstrated approach and grasp attempt but remained vulnerable to covariate shift and contact failure. Condition B was expected to begin similarly but improve after PPO refinement because reward feedback directly reinforced lift, transport, and target-region completion.

3.5.2 Hyperparameter Selection and Observations from Pilot Runs

The final hyperparameters were selected through pilot stability testing rather than exhaustive grid search. The learning rate was set to 3.0×10^{-4}; it was used for the ML-Agents trainers and selected for stable PPO refinement during pilot testing. Batch size was set to 1024 and buffer size to 10240 so that each policy update used a larger sample of the contact-rich task while preserving the standard requirement

that batch size remains smaller than buffer size. PPO clipping ϵ was set to 0.2 to limit destructive updates during refinement. Time horizon was set to 256 so that delayed grasp, lift, and transport rewards could be assigned across a longer sequence. Observation normalisation was enabled because the observation vector combines joint targets, distances, binary flags, and velocity terms at different numerical scales.

Pilot runs showed occasional reward spikes when multiple reward components triggered in the same rollout window, but the policy did not sustain stable task completion without further refinement. The release-oriented reward variant led to clear reward-hacking: the policy learned to throw or launch the cube toward the target, exploiting the absence of a post-release stability window. These observations directly informed the final reward design and the decision to report target-region success as the primary metric.

Table 3.5: Final training settings used in the implemented learning regimes. PPO-specific parameters (clip ϵ, discount γ, time horizon) apply to the PPO optimisation in Condition B (BC-assisted PPO). Condition A used ML-Agents Behavioural Cloning from the recorded demonstrations only; no extrinsic task reward contributed to the Condition A learning signal.

Parameter	Value	Rationale
Learning rate	3.0×10^{-4}	Used for the ML-Agents trainers; selected for stable PPO refinement during pilot testing.
Batch size / buffer size	1024 / 10240	Batch size remains below buffer size; larger samples reduce noisy updates.
PPO clip ϵ	0.2	Limits destructive policy updates during refinement.
Discount factor γ	0.99	Preserves credit assignment across grasp, lift, and transport phases.
Time horizon	256	Supports delayed rewards across the manipulation sequence.
Network	2 layers, 256 units	Fixed for both conditions to avoid architectural confounding.
Normalisation	Enabled	Required because observations combine quantities at different scales.
BC demonstration dataset	60 episodes	Same recorded demonstrations used for both conditions.
BC schedule	Condition A: active throughout the submitted BC-only run; Condition B: active for first 500k steps then decays to zero	Defines how demonstration loss contributes in each learning regime.

3.5.3 Reward Gating Verification

Reward-gating checks were performed through TensorBoard and visual inspection. The reward milestone sequence was designed to follow grasp, lift, transport, and target-region success. The release-oriented reward variant exposed an important reward-design failure: when release was rewarded too loosely, the policy learned to throw the object toward the target. This confirmed that stricter reward terms can create unintended strategies and that the main Condition B result should be interpreted under the target-region success criterion rather than as controlled release.

3.6 Phase III: Implemented Testing Matrix

The testing matrix covers two layers of evaluation. The first layer is the A/B training comparison between Condition A (BC-only) and Condition B (BC-assisted PPO), which establishes whether reward-based refinement improves the demonstration prior. The second layer is the robustness evaluation, which uses the converged Condition B model in inference-only mode and subjects it to eight stress conditions to assess how performance changes under controlled spatial and observation-noise perturbations.

3.6.1 A/B Training Comparison

Table 3.6: A/B training comparison matrix.

Matrix element	Condition A: BC-only	Condition B: BC-assisted PPO
Task layout	Fixed cube spawn and fixed target marker	Same fixed cube spawn and target marker
Demonstration source	Same recorded human demonstration set	Same recorded human demonstration set
Learning signal	Behavioural Cloning loss only; extrinsic reward disabled	BC assistance followed by PPO refinement with extrinsic reward enabled
Observation/action spaces	27 observations, 13 continuous actions	27 observations, 13 continuous actions
Network architecture	Two hidden layers with 256 units	Same architecture
Primary metric	Target-region success rate	Target-region success rate
Interpretation	Tests whether imitation alone is sufficient	Tests whether BC-assisted PPO refinement improves the prior

Condition A was trained using ML-Agents Behavioural Cloning from the recorded demonstrations only; no extrinsic task reward contributed to the Condition A learning signal. The submitted Condition A evidence is taken from the available ML-Agents training logs at their final reported checkpoint, approximately 1.206M logged steps. Condition B was trained using a single ML-Agents PPO trainer run in which an auxiliary BC loss was active alongside the PPO objective for the first 500,000 steps, with

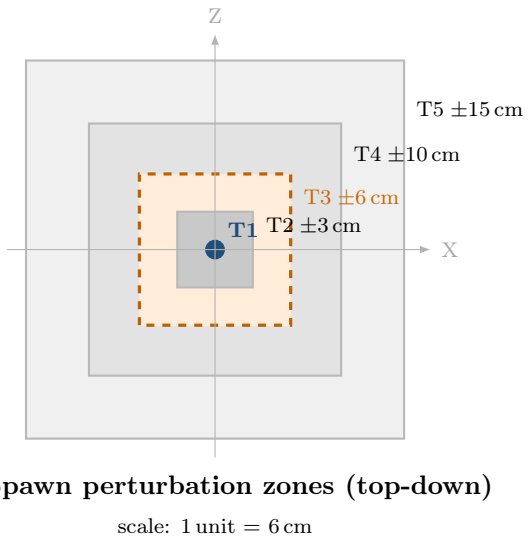
the BC loss decaying linearly to zero at that point; the PPO optimisation then continued alone for the remaining 9,500,000 steps (10,000,000 total). Both conditions used the same network architecture and observation/action spaces. The training duration differs between the two conditions because Condition B includes an extended PPO refinement phase after the early BC-assisted phase. The A/B comparison therefore evaluates the completed learning pipelines rather than an equal-budget ablation; this boundary is considered when interpreting the results.

The A/B performance comparison reports the observed training-run outcome difference between Condition A and Condition B under the fixed simulation layout. Condition A success is therefore taken from the submitted ML-Agents training logs at the final reported Condition A checkpoint (approximately 1.206M logged steps), while Condition B is supported by both training logs and the separate T1 inference baseline (100 episodes). Because Condition A was not re-evaluated through a matched held-out inference run, this comparison is interpreted as descriptive evidence about the completed pipelines rather than as a symmetrical post-training evaluation.

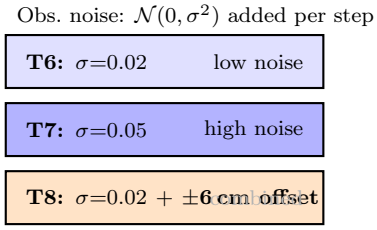
3.6.2 Robustness Evaluation: Eight-Condition Stress Test

The robustness evaluation uses the converged Condition B model and applies eight controlled perturbation conditions. No retraining takes place: the policy is fixed and evaluated in inference-only mode for **100 episodes per condition (800 episodes total)**. The conditions span spatial perturbation of the cube spawn, observation noise (additive Gaussian noise to the observation vector), and a combined perturbation-plus-noise scenario. Distractor objects are not included in this stress test because their evaluation would require additional task-specific reward design and a separate

evaluation protocol. The `StressTestController` component manages condition sequencing, applies offsets to the spawn transform and Gaussian noise to the observation vector, and writes per-episode results to a CSV file.



(a) Spawn perturbation zones (top-down)



(b) Observation noise conditions (T6–T8)

Figure 3.5: Stress-test condition overview. **(a)** Top-down view of the cube-spawn perturbation zones; each nested square is the uniform random draw region for T1–T5 (scale: 1 unit = 6 cm). The orange dashed boundary marks T3 (±6 cm), the medium spatial perturbation condition. **(b)** Observation noise profiles for T6–T8: per-step Gaussian noise $\mathcal{N}(0, \sigma^2)$ is added independently to all continuous observations (binary flags are excluded). T8 combines the T3 spatial offset with low noise.

Table 3.7: Eight-condition robustness evaluation matrix. All conditions use the fixed Condition B model.

ID	Condition name	Spawn offset	Obs noise σ	Purpose
T1	Fixed baseline	0 cm	0	Replicates the training condition; establishes unperturbed baseline
T2	Perturbation small	±3 cm	0	Minimal spatial variation
T3	Perturbation medium	±6 cm	0	Moderate spatial variation
T4	Perturbation large	±10 cm	0	Large spatial variation
T5	Perturbation XL	±15 cm	0	Near-limit spatial variation
T6	Observation noise low	0 cm	0.02	Sensitivity to low observation noise
T7	Observation noise high	0 cm	0.05	Sensitivity to high observation noise
T8	Combined	±6 cm	0.02	Spatial offset and noise together

The spawn offset is applied as a uniform random draw in $[-r, r]$ on both the X and Z axes relative to the default spawn position. The observation noise is sampled from $\mathcal{N}(0, \sigma^2)$ and added independently to each continuous observation at every decision step. The noise is not applied to binary flags such as the grasp-validation signal. Tasks such as object throwing, bouncing, object classification, and stacking are excluded because each would require a different reward function, new success criteria, and additional demonstrations.

The stress-test analysis examines how success rate and supporting failure met-

rics change relative to the fixed-layout baseline under controlled spatial offsets, additive observation noise, and a combined perturbation condition.

3.7 Evaluation Framework and Statistical Analysis

3.7.1 Performance Metrics

The primary metric is task success rate, defined as the proportion of evaluation episodes in which the cube enters the target-region radius (0.15 m). Mean time-to-success is computed only over successful episodes and is measured from episode start until the cube first enters the target-region radius. Supporting metrics include proxy grasp-validation rate, physical lift rate, drop rate, timeout rate, and minimum distance to the target. These supporting metrics are required because a binary success rate alone can hide meaningful partial progress, especially in contact-rich manipulation. Training efficiency is reported descriptively using total training steps, TensorBoard learning trends, and the checkpoint at which stable target-region success is first observed.

Runtime feasibility is assessed descriptively during Unity inference-mode evaluation. The system is considered near-real-time if policy decisions are applied continuously during simulation without visible control stalls, frame blocking, or delayed action updates. This is not treated as a hardware-independent latency benchmark because runtime performance depends on the host machine, Unity frame rate, and scene configuration. In Chapter 4 the runtime behaviour is reported descriptively based on observed inference performance.

3.7.2 Proxy Grasp Reliability Analysis

The GraspValidator subsystem is analysed as a proxy event rather than as a ground-truth grasp detector. No independent manually labelled grasp dataset was collected, so formal precision, recall, and F1 scores cannot be computed against an external ground truth. Instead, a validated grasp is treated as *reliable* when it is followed by a physical lift or eventual target-region success, and *unreliable* when it is followed by a drop, timeout, or no lift. These proportions are reported as decomposed reliability indicators (e.g., the proportion of validated grasps that lead to lift) rather than as a formal classifier confusion matrix. This distinction prevents the dissertation from equating an internal grasp flag with real physical manipulation success.

3.7.3 Descriptive Analysis Protocol

The completed evaluation is analysed using descriptive statistics. For each trained policy, task success is reported as the proportion of evaluation episodes in which the cube enters the target-region radius. Mean time-to-success is computed only over successful episodes. Supporting metrics include proxy grasp-validation rate, physical lift rate, drop rate, timeout rate, and minimum distance to the target.

The A/B comparison reports the observed training-run outcome difference between Condition A and Condition B under the fixed simulation layout. Condition A success is taken from the submitted ML-Agents training logs at the final reported Condition A checkpoint (approximately 1.206M logged steps), while Condition B is supported by both training logs and the separate T1 inference baseline (100 episodes). Because Condition A was not re-evaluated through a matched held-out inference run, this comparison is interpreted as descriptive evidence about the completed pipelines

rather than as a symmetrical post-training evaluation.

The robustness evaluation treats the converged Condition B policy as a fixed inference model. Each stress-test condition is summarised using success rate, performance change relative to the fixed-layout baseline, and supporting failure metrics. The 800 stress-test episodes estimate the robustness of this trained policy under spatial and observation-noise perturbations; they are not treated as substitutes for multiple independently trained policies.

3.8 Ethical and Validity Constraints

3.8.1 Ethical Data Protocol

The dataset consists of numerical robot observations and actions recorded from the researcher's own demonstrations. The capture pipeline does not require storing identifiable video in the final training dataset. The retained data are anonymised numerical state-action pairs and training logs.

3.8.2 Internal and External Validity

Internal validity is supported by the fixed scene layout and shared dataset between conditions. External validity is limited because the work is simulation-only and uses one operator. The results therefore support claims about the implemented Unity prototype and its BC versus BC-assisted PPO comparison, not direct transfer to a physical robot or general dexterous manipulation across arbitrary tasks.

3.9 Chapter Summary

This chapter specified the controlled fixed-scene A/B methodology and the separate inference-time robustness evaluation used to assess the completed BC-only and BC-assisted PPO pipelines. The study compares a BC-only baseline against a BC-assisted PPO refinement policy using the same demonstration set, robot, scene, observation space, and action space, while reporting trainer-specific settings separately. Condition B is implemented as a single PPO training run in which an auxiliary BC loss is active during the early phase and decays to zero, after which PPO continues alone. The final implementation uses 27 observations and 13 continuous actions. The main evaluation measures target-region success rather than controlled release, and the attempted release-oriented variant is treated as a reward-design limitation rather than the primary result. The chapter also defines the descriptive analysis boundary. The A/B comparison reports observed training-run outcomes between one trained policy per condition, while the robustness test evaluates one fixed Condition B policy under inference-time perturbations. The next chapter presents the results of the implemented A/B comparison and the eight-condition robustness stress test.

Chapter 4

Analysis of Results and Discussion

4.1 Chapter Overview

This chapter presents the results of the completed A/B training comparison and the separate inference-time robustness evaluation defined in Chapter 3. The A/B comparison examines whether Behavioural Cloning (BC) alone was sufficient for the implemented manipulation task, and whether BC-assisted Proximal Policy Optimisation (PPO) refinement improved the same demonstration-based policy pipeline. The robustness evaluation then tests the converged Condition B policy under eight controlled perturbation conditions, without retraining, to assess whether the learned behaviour remains reliable beyond the fixed training layout.

The central result is two-sided. Under the fixed-scene A/B comparison, Condition A remained at 0% target-region success during the submitted BC-only training run, up to approximately 1.206M logged steps, while Condition B reached approximately 95% target-region success after PPO refinement. This supports the dissertation hypothesis within the implemented simulation boundary: captured human-motion demonstrations can provide a useful behavioural starting point, but reward-based refinement is required for reliable target-region task completion. However, the

robustness stress test shows that this success does not generalise equally across all perturbations. The final Condition B policy achieved 100% success under the fixed baseline and under additive observation noise alone, but success fell sharply under cube-spawn perturbation, reaching 19% at ±6 cm and 23% when ±6 cm perturbation was combined with low observation noise.

The success criterion must therefore be interpreted precisely. The reported success rate measures whether the cube enters the target-region radius. It does not prove stable release after hand opening, nor does it prove robust generalisation to arbitrary object placements. The valid claim is simulated grasp-and-transport-to-target behaviour under the implemented target-region criterion, with strong fixed-layout performance but weak spatial robustness.

4.2 Relationship to the Implemented Evaluation Matrix

Chapter 3 defined two layers of evaluation. The first layer is the A/B training comparison between Condition A: BC-only and Condition B: BC-assisted PPO refinement. The second layer is the robustness stress test, which evaluates the converged Condition B policy only. These two layers answer different questions and must not be collapsed into one result. The A/B comparison asks whether reward-based refinement improves the completed learning pipeline. The stress test asks whether the final refined policy remains reliable when the cube spawn or observations are perturbed.

Table 4.1: Results chapter alignment with the implemented evaluation matrix.

Evaluation layer	Evidence used	Interpretation boundary
A/B training comparison	Condition A training logs to approximately 1.206M logged steps and Condition B training logs after PPO refinement	Descriptive comparison of completed learning pipelines, not a matched multi-seed statistical test
Fixed-layout inference baseline	T1 stress-test baseline for final Condition B policy, 100 episodes	Confirms fixed-layout inference performance for Condition B only
Robustness stress test	T1–T8, 100 episodes per condition, final Condition B policy only	Estimates robustness of one trained policy under inference-time perturbations; not a substitute for multiple training seeds

This distinction is important because Condition A was not re-evaluated through a matched held-out inference run. Its result is taken from the completed BC-only training logs. Condition B is supported by both training logs and the separate T1 inference baseline. The A/B result is therefore reported as a training-run outcome comparison, while the stress-test result is reported as an inference-time robustness analysis of the final Condition B policy.

4.3 Training Data and Demonstration Boundary

Both conditions used the same recorded human-demonstration dataset. The final methodology defines this dataset as 60 recorded demonstration episodes. These demonstrations were collected through the Kinect–MediaPipe capture pipeline and

recorded by Unity ML-Agents as state-action examples. The demonstrations provide the common behavioural prior for the two conditions: in Condition A they are the only learning signal, while in Condition B they provide early BC assistance before PPO refinement becomes dominant.

The dataset size frames the work as a small-demonstration R2S study rather than a large-scale imitation-learning benchmark. The purpose of the demonstrations is therefore not to cover every possible contact state, object displacement, or recovery scenario. Their role is to encode the human task sequence sufficiently well that the policy begins near useful behaviour. The later results show that this was not enough for BC-only success, but it was enough to support PPO refinement.

4.4 Condition A: Behavioural Cloning Only

Condition A trained the policy using Behavioural Cloning as the only learning signal. No extrinsic reward contributed to the Condition A learning signal. The policy could therefore only minimise action mismatch against the demonstration distribution; it could not receive feedback for grasp validation, lift, target-region approach, drop events, or eventual success.

The BC-only run did not solve the implemented task. At the final reported Condition A checkpoint, approximately 1.206M logged steps, target-region success remained 0%. This does not mean the policy learned nothing. The training logs show that by the final checkpoint the closure validator was triggering consistently (closure-validator rate: 1.0), indicating that BC did learn the hand-closure component of the task. However, the dominant episode outcome was a drop rather than success or timeout: the drop count reached approximately 7 per 10-episode logging window

while the timeout rate remained at 0%. This shows that the BC policy could close the hand, but could not sustain the grasp through the transport phase. The failure is therefore not a complete absence of learning; it is a partial reproduction that breaks down precisely at the contact-rich transition from grasp initiation to object transport.

Table 4.2: Condition A training outcome summary.

Item	Observed outcome
Learning signal	Behavioural Cloning from demonstrations only
Training evidence	Submitted ML-Agents logs up to approximately 1.206M logged steps
Target-region success	0% at the final reported checkpoint
Observed behaviour	Approach and grasp-attempt behaviour, but no reliable lift-transport-target completion
Main failure interpretation	Demonstration matching alone did not provide recovery from contact errors or unseen execution states
Closure-validator rate	1.0 at convergence in the logged episode window
Dominant episode outcome	Drop; approx. 7 per 10-episode logging window; timeout rate 0%
Mean episode length	Approx. 132 decision steps at convergence

The Condition A result supports the literature-based concern that BC is fragile under covariate shift in dexterous manipulation. A demonstration-only policy can reproduce states that look similar to the data, but it is not trained to correct its own mistakes after the object moves unexpectedly, the fingers close slightly off-position, or the cube fails to lift. For this reason, the 0% result is not merely a failed baseline; it identifies the missing role of reward-based interaction.

4.5 Condition B: BC-Assisted PPO Refinement

Condition B used the same demonstration dataset, but the training regime differed. The policy was trained using a PPO run in which auxiliary BC loss was active during the early phase and decayed to zero after 500,000 steps. PPO then continued alone for the remaining training budget. This allowed the policy to begin from demonstration-assisted behaviour while later receiving direct reward feedback for grasp, lift, transport progress, and target-region completion.

The final Condition B training run reached approximately 95% target-region success after PPO refinement. The T1 fixed-layout inference baseline later achieved 100% success over 100 episodes. These two values should not be treated as contradictory: the 95% value summarises the converged training outcome, while the 100% T1 value is a separate fixed-layout inference measurement of the final checkpoint. Together, they show that the final refined policy solved the fixed-layout target-region task reliably.

Table 4.3: Condition B training and fixed-layout inference summary.

Item	Observed outcome
Learning signal	Early BC assistance followed by PPO-only refinement
Training duration	10,000,000 reported ML-Agents training steps
Training success outcome	Approximately 95% target-region success after convergence
T1 fixed-layout inference	100 successful episodes out of 100
Mean T1 time-to-success	0.762 s over successful T1 episodes
Main interpretation	Reward refinement converted the demonstration prior into reliable fixed-layout target-region completion
Closure-validator rate	1.0 at convergence in the logged episode window
Dominant episode outcome	Success; drop count approx. 1 per 10-episode logging window; timeout rate 0%
Mean episode length	Approx. 77 decision steps at convergence

The improvement from Condition A to Condition B should not be interpreted as "BC failed and PPO succeeded" in a simplistic sense. A stronger interpretation is that the demonstration data supplied task direction, but did not contain enough recovery information to solve contact-rich execution by imitation alone. PPO supplied the missing corrective mechanism by rewarding the policy under its own execution distribution.

4.6 A/B Comparison and Refinement Gain

The A/B comparison shows a large descriptive separation between the completed pipelines. Condition A remained at 0% target-region success, while Condition B

reached high fixed-layout success after PPO refinement. Because the comparison uses one completed training run per condition and unequal training budgets, the result is not presented as a compute-controlled or statistically replicated ablation. It is instead interpreted as descriptive evidence that, in the implemented R2S pipeline, imitation alone was insufficient and reward refinement was necessary for reliable target-region task completion.

Table 4.4: High-level comparison of the completed learning pipelines.

Metric / observation	Condition A: BC-only	Condition B: BC-assisted PPO
Training signal	Demonstration action matching only	Demonstration assistance plus PPO reward refinement
Final training outcome	0% target-region success at approximately 1.206M logged steps	Approximately 95% target-region success after PPO convergence
Fixed-layout inference	Not run as a matched held-out test	100% success in T1 baseline, 100 episodes
Closure-validator rate	1.0 at convergence in the logged episode window	1.0 at convergence in the logged episode window
Dominant episode outcome	Drop (approx. 7 per 10-ep window); timeout 0%	Success (approx. 1 drop per 10-ep window); timeout 0%
Mean episode length	Approx. 132 decision steps	Approx. 77 decision steps
Behavioural interpretation	Partial reproduction of approach/grasp attempt, no reliable task completion	Reliable fixed-layout grasp-and-transport-to-target behaviour
Main weakness	No correction mechanism after contact errors	Strong fixed-layout performance but weak spatial robustness
Claim boundary	Demonstration-only learning was insufficient in this implementation	Refinement solved the fixed-layout target-region task, not full general manipulation

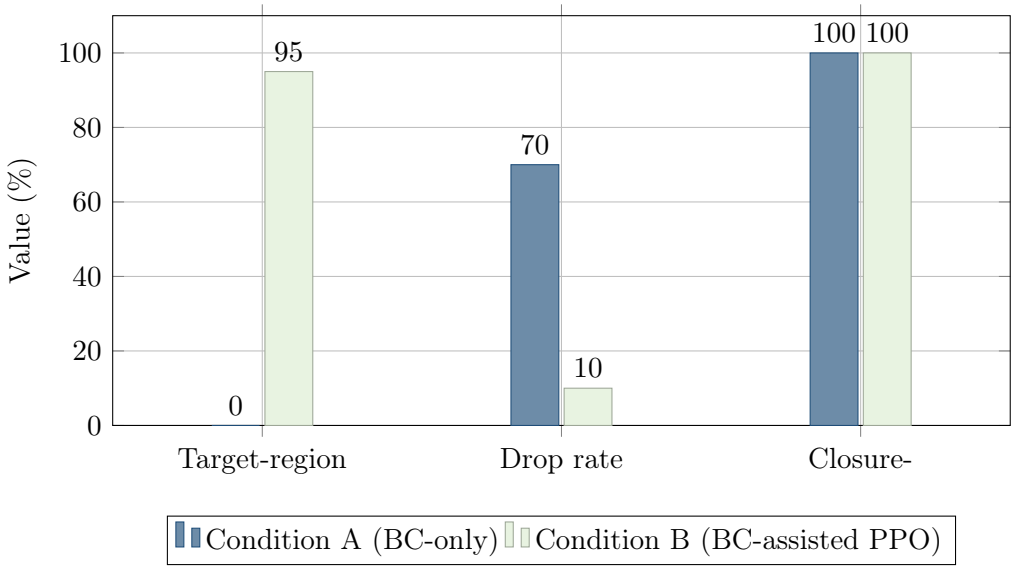


Figure 4.1: Comparison of key converged training metrics for Condition A and Condition B. Both conditions reached a closure-validator rate of 100%, indicating that hand closure was learned under both regimes. The critical separation is in post-closure outcomes: Condition A produced drops in approximately 70% of logged episodes and 0% target-region success, while Condition B reduced the drop rate to approximately 10% and reached approximately 95% target-region success. Drop rate is an approximate percentage derived from the ML-Agents 10-episode logging window at convergence; closure-validator rate and target-region success are taken from the final training checkpoint.

The training metrics add important nuance to this comparison. Both conditions reached a closure-validator rate of 1.0 at convergence, indicating that the hand-closure component of the task was learned under both regimes. The critical difference lies in what followed closure: in Condition A the dominant episode outcome was a drop (approximately 7 per 10-episode logging window, timeout rate 0%), whereas in Condition B drops became rare (approximately 1 per 10-episode logging window) and success dominated. This shows that BC reproduced grasp initiation but failed during post-grasp transport, which is the contact-rich phase where covariate shift is most damaging. PPO refinement primarily improved post-grasp transport and recovery rather than merely teaching hand closure. The observed difference is therefore practically meaningful, but it must be framed correctly: the result supports the value of reward-based refinement for post-grasp transport in this implementation; it does not

establish a general algorithmic superiority claim across random seeds, embodiments, tasks, or object layouts.

4.7 Robustness Evaluation of the Final Condition B Policy

The eight-condition stress test evaluates the final Condition B policy under inference-time perturbations. No retraining occurs during this evaluation. Each condition contains 100 episodes, giving 800 evaluation episodes in total. The test includes a fixed baseline, four cube-spawn perturbation levels, two observation-noise levels, and one combined spatial-plus-noise condition.

Table 4.5: Condition B robustness stress-test results ($n = 100$ episodes per condition). Drop = episode ended without success and without timeout. Time-to-success is computed only over successful episodes.

ID	Spawn offset	Obs. noise σ	Success	Drop	Timeout	Mean TTS	Mean min. dist.
T1	0 cm	0.00	100%	0%	0%	0.762 s	0.139 m
T2	±3 cm	0.00	69%	27%	4%	0.794 s	0.394 m
T3	±6 cm	0.00	19%	60%	21%	0.789 s	0.936 m
T4	±10 cm	0.00	5%	88%	7%	0.820 s	1.138 m
T5	±15 cm	0.00	9%	87%	4%	0.818 s	1.131 m
T6	0 cm	0.02	100%	0%	0%	0.764 s	0.138 m
T7	0 cm	0.05	100%	0%	0%	0.769 s	0.138 m
T8	±6 cm	0.02	23%	58%	19%	0.813 s	0.893 m

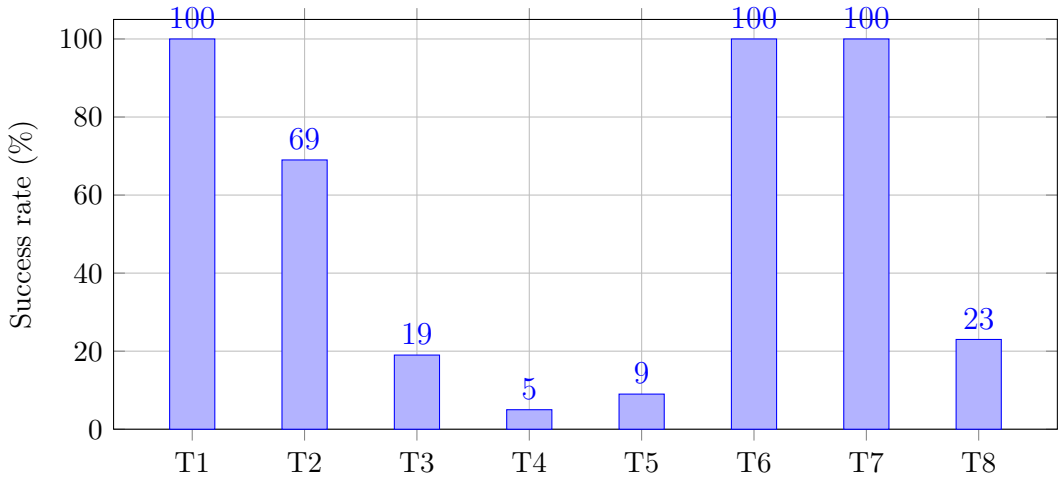


Figure 4.2: Condition B stress-test success rates. The final policy achieves 100% success under fixed-layout and observation-noise-only conditions, but fails sharply under spatial spawn perturbation.

The stress-test results are the most important addition to the evaluation because they prevent the dissertation from overclaiming the 95% fixed-layout result. Under T1, the final policy succeeded in all 100 fixed-layout inference episodes. Under T6 and T7, additive Gaussian observation noise alone does not reduce success. In contrast, spatial perturbation of the cube spawn produces a steep drop: 69% at $\pm 3\,\text{cm}$, 19% at $\pm 6\,\text{cm}$, and near-floor performance at $\pm 10\,\text{cm}$ and $\pm 15\,\text{cm}$. T5 is slightly higher than T4, but both values are so low that this should not be interpreted as recovery at larger offsets; with 100 episodes per condition, the correct interpretation is that both large-offset conditions are failure-dominated.

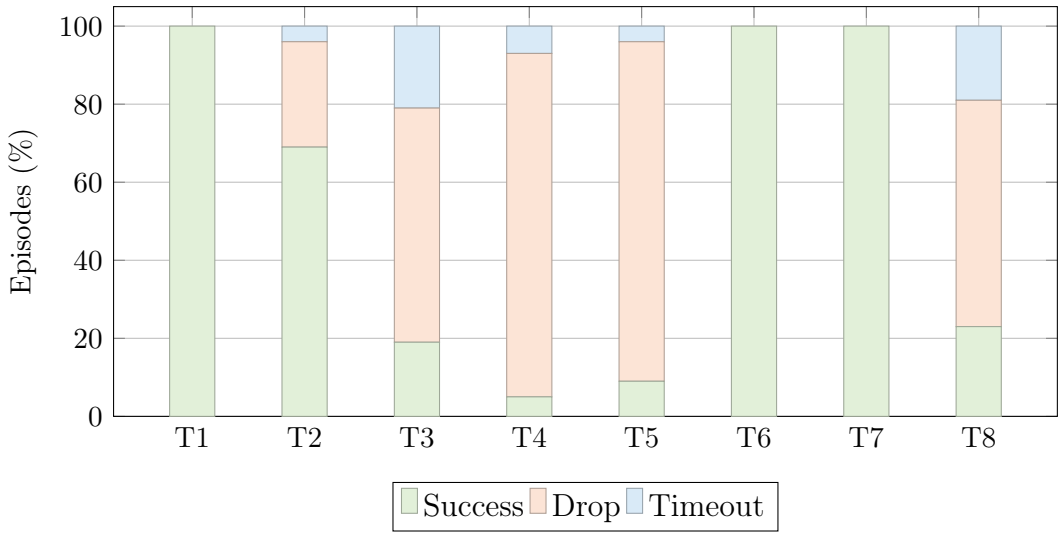


Figure 4.3: Episode outcome breakdown across all eight stress-test conditions. Each bar sums to 100%. Success dominates under fixed-layout and noise-only conditions; drop rate rises sharply with spatial perturbation while timeout remains comparatively low.

4.8 Interpretation of Spatial Robustness

The spatial perturbation results show that the learned behaviour is position-specific. The policy is not merely learning a general rule such as "find cube, grasp cube, move cube to target" across a wide workspace. Instead, it appears to rely heavily on the object geometry and motion sequence encountered during fixed-layout training. A small $\pm 3\,\text{cm}$ perturbation remains partly recoverable, but by $\pm 6\,\text{cm}$ the policy fails in most episodes.

The minimum-distance values support this interpretation. Successful episodes naturally end near the 0.15 m target-region threshold, while failed spatial-perturbation episodes often remain far from the target. Mean minimum distance rises from 0.139 m in T1 to 0.936 m in T3, 1.138 m in T4, and 1.131 m in T5. This means the policy is not merely missing the target by a small margin under large offsets. In many failed episodes, the object does not approach the target region at all.

The result directly qualifies the fixed-layout success claim. PPO refinement

solved the trained scene, but did not produce broad spatial generalisation. This is not a minor limitation; it is a central finding of the stress test. The trained policy is robust to noisy observations of a familiar layout, but brittle when the initial object placement changes.

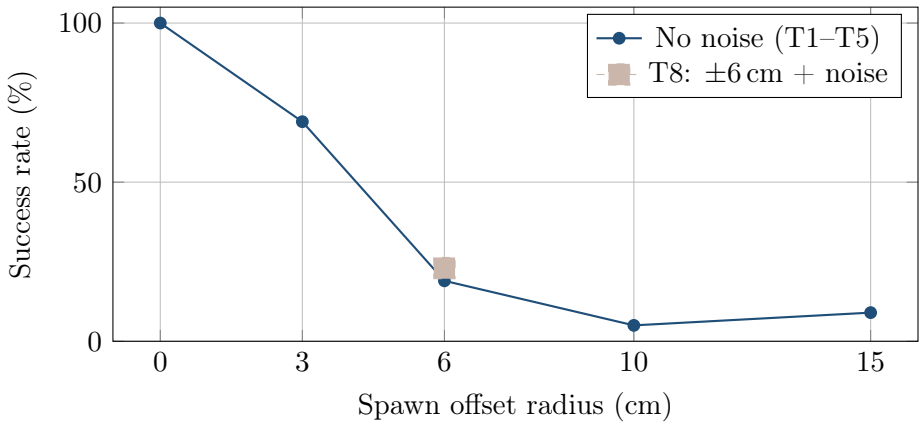


Figure 4.4: Success rate as a function of cube-spawn offset radius for noise-free conditions (T1–T5). T8 (combined ±6 cm with low observation noise) is shown as a separate marker for comparison. Performance collapses sharply beyond ±3 cm, and the addition of noise at the same offset does not materially worsen the outcome.

4.9 Observation Noise and Combined Perturbation

Observation noise alone did not reduce success. T6 and T7 both achieved 100% success, with mean time-to-success remaining close to the fixed baseline: 0.764 s for T6 and 0.769 s for T7, compared with 0.762 s for T1. This indicates that the final policy can tolerate the tested levels of additive observation noise when the cube remains at the trained spawn position.

The combined condition changes the interpretation. T8 combines the ±6 cm spatial offset with low observation noise and achieved 23% success. This is close to the 19% success observed in T3, which used the same spatial offset without noise. Therefore, the dominant failure driver appears to be the spatial offset rather than the tested observation noise. The low-noise addition did not collapse an otherwise robust

policy; the policy was already brittle under the 6 cm spatial displacement.

This distinction matters for future design. Improving sensor filtering alone would not solve the main robustness weakness. The next technical priority would be training-time spatial randomisation, curriculum learning, or a reward/observation design that encourages object-relative rather than spawn-specific control.

4.10 Efficiency and Time-to-Success

Time-to-success is computed only over successful episodes. Under fixed-layout and noise-only conditions, completion was fast and stable: 0.762 s for T1, 0.764 s for T6, and 0.769 s for T7. Under spatial perturbation, successful episodes took slightly longer: 0.794 s for T2, 0.789 s for T3, 0.820 s for T4, 0.818 s for T5, and 0.813 s for T8. The increase is modest among successful episodes, but this metric must be interpreted alongside success rate. A low time-to-success in T3 or T8 does not mean the condition is robust, because most episodes in those conditions failed.

The results therefore support RQ3 only within the correct boundary. Observed Unity inference behaviour supports near-real-time use inside the simulation, but this is reported descriptively rather than as a hardware-independent latency benchmark. Successful episodes complete quickly. Training efficiency is weaker: Condition B required an extended PPO refinement run, and Condition A failed after the completed BC-only run. The system is therefore efficient at inference once trained, but not cheap to train.

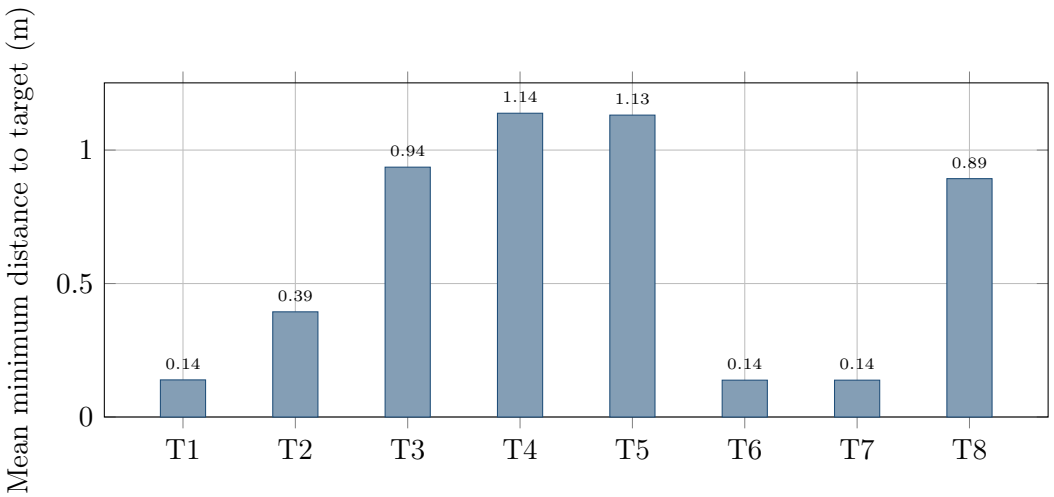


Figure 4.5: Mean minimum distance of the object to the target region across all stress-test conditions. Conditions T1, T6, and T7 achieve proximity near the 0.15 m success threshold. Under large spatial offsets (T3–T5, T8) the object frequently fails to approach the target region at all, confirming that failure is not a near-miss but a full task breakdown.

4.11 Comparison with Related Work

The A/B result aligns with the pattern reported in the dexterous-manipulation literature reviewed in Chapter 2. The key empirical pattern is that demonstration-only learning is usually weakest in contact-rich manipulation, while refinement through reinforcement learning produces the major performance gain. DIME reported 0% BC success across four dexterous tasks, while later improvement reached up to 75% success [8]. RialTo similarly reported weak BC-only performance of 5–25%, increasing to 75–91% after refinement [7]. DexTransfer reported an improvement from 11% BC-only success to 61–74% after refinement, while DexMV showed that vision-derived human demonstrations can substantially improve dexterous robot learning [21, 9].

The present result follows the same direction. Condition A remained at 0% target-region success, while Condition B reached approximately 95% target-region success after BC-assisted PPO refinement. This places the dissertation within the same empirical pattern as DIME, RialTo, DexTransfer, and DexMV: demonstrations

are useful, but demonstration matching alone is fragile when the task requires stable contact, lift, and post-grasp transport. The comparison is especially important because Condition A did not fail to learn all behaviour. It reached a closure-validator rate of 1.0, indicating that the hand-closure component was learned, but the policy still dropped the cube during transport. The result therefore refines the literature comparison: in this implementation, BC was sufficient for grasp initiation but insufficient for post-grasp control.

Table 4.6: Relationship between this dissertation and representative empirical literature reviewed in Chapter 2.

Study	Learning pattern	Reported outcome	Relation to this dissertation
DIME [8]	BC fragility in dexterous manipulation	0% BC, up to 75% after improvement	Supports the 0% BC baseline as plausible in contact-rich dexterous tasks
RialTo [7]	Demonstrations refined through RL in real-to-sim-to-real tasks	5–25% BC, 75–91% after refinement	Similar BC-to-refinement improvement pattern, but with stronger real-world placement/release endpoint
DexTransfer [21]	Few MoCap demonstrations plus refinement	11% BC, 61–74% after refinement	Shows that higher-fidelity demonstrations can produce stronger BC starts than low-cost vision capture
DexMV [9]	Vision-derived human demonstration transfer	Demonstrations improve dexterous robot learning	Closest conceptual link to vision-based demonstration transfer, but differs in task and evaluation setup
This dissertation	Low-cost R2S demonstrations with BC-assisted PPO	0% BC, approx. 95% fixed-layout target-region success after refinement	Confirms the refinement-gain pattern, while stress testing shows weak spatial robustness

The comparison strengthens the interpretation of the A/B result, but it also prevents overclaiming. Numerically, the movement from 0% BC success to approxi-

mately 95% fixed-layout success is larger than several reported gains in the reviewed literature. However, this does not mean the present system is stronger than those systems. The endpoint is easier to satisfy because success is defined by target-region entry in simulation rather than stable release, real-world placement, or physical robot deployment. RialTo, for example, evaluates real-world manipulation robustness, while this dissertation evaluates a controlled simulation task [7]. DexTransfer uses motion-capture demonstrations, while the present work uses a lower-cost vision-based pipeline [21]. Therefore, the correct comparison is directional rather than absolute: this dissertation reproduces the same empirical pattern that BC alone is fragile and RL refinement improves task completion, but under a narrower and more controlled success criterion.

The robustness results also add a qualification that many fixed-condition success tables can hide. The final Condition B policy achieved 100% success in the fixed-layout T1 baseline and in both observation-noise-only conditions, but success dropped sharply under cube-spawn perturbation. This means the policy learned the trained spatial arrangement more strongly than a general object-relative manipulation strategy. Related work such as RialTo frames refinement as a way to improve robustness to object poses, disturbances, and visual changes [7]. In contrast, the present stress test shows that although PPO refinement solved the fixed scene, it did not produce broad spatial robustness. This difference is not a weakness in reporting; it is one of the main empirical findings of the dissertation.

4.12 Statistical and Replication Boundary

The completed evaluation is descriptive. Only one final training run was completed per condition, so inferential statistical tests across independent seeds are not reported. A p-value from one seed per condition would be misleading because it would not estimate training-run variability. Deep reinforcement learning results are known to be sensitive to implementation detail, random seed, environment nondeterminism, and reporting choices; Henderson et al. [36] specifically warn that results can be difficult to interpret without careful reporting and significance-aware evaluation. Colas et al. [37] similarly argue that the number of random seeds directly affects the reliability of statistical conclusions in deep reinforcement learning experiments.

Table 4.7: Statistical interpretation boundary for the completed results.

Evidence type	**Interpretation**
A/B training result	Descriptive comparison of one completed Condition A run and one completed Condition B run
Stress-test episodes	800 inference episodes for one fixed Condition B policy
Inferential statistics	Not reported because multiple independent training seeds were not completed
Valid claim	Strong internal evidence for this implementation, not general algorithm-level proof

This boundary weakens generalisability, but it makes the reporting more credible. The observed separation between Condition A and Condition B is large: Condition A remained at 0% target-region success, while Condition B reached approximately 95% target-region success after refinement and achieved 100 successful episodes out of

100 in the T1 fixed-layout inference baseline. However, this should be interpreted as a completed-run outcome comparison rather than a statistically replicated algorithm comparison.

The 800 stress-test episodes strengthen a different part of the evaluation. They provide a more detailed inference-time analysis of one trained Condition B policy by testing fixed-layout behaviour, spatial perturbation, observation noise, and combined perturbation. They do not replace multiple independently trained policies. In other words, the stress test supports claims about the robustness profile of the final trained policy, while multiple random seeds would be required to support claims about the expected training behaviour of the learning algorithm.

The final statistical position is therefore deliberately conservative. The dissertation can claim strong internal evidence that BC-assisted PPO refinement improved the implemented pipeline and that the final policy was brittle under spatial perturbation. It cannot claim that BC-assisted PPO is generally superior across all random seeds, task layouts, embodiments, or capture systems. This distinction keeps the results aligned with best practice in deep reinforcement learning reporting [36, 37].

4.13 Discussion Against the Research Questions

4.13.1 RQ1: Can the virtual arm and hand accomplish the manipulation sequence?

The answer is yes, but only under Condition B and only under the implemented success criterion. BC-only learning did not accomplish the sequence because it remained at 0% target-region success during the submitted Condition A training run, up to approximately 1.206M logged steps. However, the training logs show that Condition A

did reach a closure-validator rate of 1.0, meaning the failure was not a complete absence of learned task structure. The BC-only policy learned hand closure but failed during post-grasp transport, where drops dominated the episode outcomes.

BC-assisted PPO did accomplish the fixed-layout target-region task, reaching approximately 95% target-region success in training and 100% success in the T1 fixed-layout inference baseline. Empirically, this movement from 0% BC success to high PPO-refined success matches the refinement-gain pattern reported in DIME, RialTo, DexTransfer, and DexMV [8, 7, 21, 9]. The result therefore supports RQ1 for the BC-assisted PPO condition, but not for imitation alone.

The answer must remain bounded. The system can perform simulated grasp-and-transport-to-target behaviour in the trained layout, but it should not be described as full autonomous pick-place-release because controlled release is not enforced. The empirical result is therefore "BC-only: closure without successful transport" to "BC-assisted PPO: reliable fixed-layout target-region transport", not "human-level dexterous manipulation".

4.13.2 RQ2: Does the learned policy reproduce the task-level structure of demonstrations?

Condition B reproduces the task-level structure of the demonstrations at the level measured in this dissertation: approach, hand closure, object movement, transport, and target-region completion. This is consistent with vision- and demonstration-based dexterous manipulation studies such as DexMV, where human demonstration signals are used to transfer task structure into robot learning [9]. In the present work, the important result is that the demonstration signal was not useless: it helped initialise

behaviour that PPO could later refine.

However, the stress test shows that this task-level reproduction was position-specific. The policy behaved reliably when the cube started in the trained location, but success declined sharply when the cube spawn was perturbed. The empirical movement is clear: T1 fixed-layout success was 100%, but success fell to 69% at ±3 cm, 19% at ±6 cm, 5% at ±10 cm, 9% at ±15 cm, and 23% under the combined ±6 cm plus low-noise condition. This means the policy reproduced the demonstrated task sequence in the familiar layout, but did not learn broad object-location generalisation.

Compared with related work, this is a narrower result than systems that explicitly aim to robustify policies to object pose variation or real-world environmental disturbance [7]. The present study therefore answers RQ2 positively only within the fixed-layout simulation boundary. The policy learned a task-level sequence, but the robustness evaluation shows that the sequence was not yet generalised into an object-relative manipulation strategy.

4.13.3 RQ3: Is the system efficient and near-real-time?

The trained policy is efficient at inference but not efficient to train. Successful Condition B episodes complete in less than one second in the Unity simulation under the measured successful cases, and the policy runs through the Unity decision loop without offline planning. This supports the near-real-time runtime component of RQ3 inside the implemented simulation environment.

Training efficiency is weaker. Condition A failed during the submitted BC-only training run, and Condition B required extended PPO refinement before reaching stable target-region success. This reflects the broader trade-off in the reviewed literature:

reinforcement learning can improve policies through interaction, but refinement is not free and can require substantial training time [7, 36]. The present result therefore supports runtime efficiency after training, not lightweight training efficiency.

The final answer to RQ3 is therefore split. At runtime, the final policy is suitable for near-real-time simulated control within the tested Unity environment. During training, the pipeline remains computationally expensive and should be treated as an offline learning process. A stronger claim about real-time deployment would require hardware-independent latency measurements, repeated runs, and physical deployment testing, none of which were part of the completed evaluation.

4.14 Main Limitations Identified by the Results

The first limitation is the success metric. Target-region entry does not equal stable release. The results support grasp-and-transport-to-target behaviour, not full pick-place-release. This is especially important because the simplified GraspValidator is only a closure-stability proxy.

4.14.1 Proxy Interpretation of Grasp-Validation Outcomes

The GraspValidator output is not treated as a ground-truth grasp label. A formal confusion matrix with precision, recall, and F1 score would require an independently labelled dataset identifying whether each episode contained a physically valid grasp. Such a dataset was not collected. Therefore, the validator is interpreted as an internal proxy signal and analysed against later task outcomes.

For interpretation, the GraspValidator output is discussed using true-positive-like, false-positive-like, true-negative-like, and false-negative-like categories. A true-

positive-like event occurs when the validator triggers and the episode later produces a physical lift or target-region success. A false-positive-like event occurs when the validator triggers but the episode later drops, times out, or fails to lift, meaning that the closure-based signal did not lead to meaningful manipulation. A true-negative-like event occurs when the validator does not trigger and the episode also fails to lift or reach the target. A false-negative-like event occurs when the validator does not trigger but the episode still produces lift or target-region success.

These labels are used as proxy interpretations only. They are not reported as formal classifier true positives, true negatives, false positives, false negatives, precision, recall, or F1 score, because no independent manually labelled grasp dataset was collected. The purpose is to discuss whether the internal validator signal aligns with later task behaviour, not to validate it as a ground-truth grasp detector.

Table 4.8: Proxy true-positive-like, false-positive-like, true-negative-like, and false-negative-like interpretation categories for GraspValidator outcomes. Categories are descriptive only; no ground-truth grasp labels were independently collected.

Validator output	Later episode outcome	Interpretation
Triggered	Physical lift or target-region success	True-positive-like proxy event
Triggered	Drop, timeout, or no lift	False-positive-like proxy event
Not triggered	No lift and no target-region success	True-negative-like proxy event
Not triggered	Lift or target-region success	False-negative-like proxy event

This proxy framing matters because the dissertation's main success metric is target-region completion rather than independently verified stable release. It prevents

the internal grasp flag from being treated as proof of a physically valid grasp and keeps the 95% fixed-layout success result bounded to simulated grasp-and-transport-to-target behaviour.

4.14.2 Proxy Precision, Recall, and Operating Curve Across Test Conditions

Because the closure-validator triggered in every evaluated episode under Condition B, proxy recall is approximately 1.0 across all eight test conditions: every successful episode was preceded by a validator trigger, so there are no false-negative-like events. The discriminating metric is therefore proxy precision, which is equal to the per-condition success rate. Table 4.9 reports proxy precision, recall, and F1 score for each condition.

Table 4.9: Proxy precision, recall, and F1 score per stress-test condition (100 episodes each), derived from the closure-validator proxy framing. Recall is approximately 1.00 in all conditions because the validator fired in every episode. Precision tracks the per-condition success rate and degrades with perturbation severity.

Cond.	Description	Episodes	Successe	Proxy prec.	Proxy rec.	Proxy F1
T1	Fixed layout	100	100	1.00	≈1.00	1.00
T2	±3 cm offset	100	69	0.69	≈1.00	0.82
T3	±6 cm offset	100	19	0.19	≈1.00	0.32
T4	±10 cm offset	100	5	0.05	≈1.00	0.10
T5	±15 cm offset	100	9	0.09	≈1.00	0.17
T6	Fixed layout + low noise	100	100	1.00	≈1.00	1.00
T7	Fixed layout + medium noise	100	100	1.00	≈1.00	1.00
T8	±6 cm + low noise	100	23	0.23	≈1.00	0.37

Precision is 1.00 in the fixed-layout conditions (T1, T6, T7), declines to 0.69 under small spatial perturbation (T2), and falls to 0.05 and 0.09 under larger offsets (T4, T5). Because recall remains near 1.00 throughout, the F1 score tracks precision closely; F1 drops from 1.00 to 0.10 at peak degradation. The slight recovery from T4 to T5 is an artefact of very large offsets placing the cube near environmental reach limits, occasionally producing accidental successes. The combined condition T8 produces lower precision than the noise-only conditions T6 and T7, confirming that spatial variation is the dominant degradation factor.

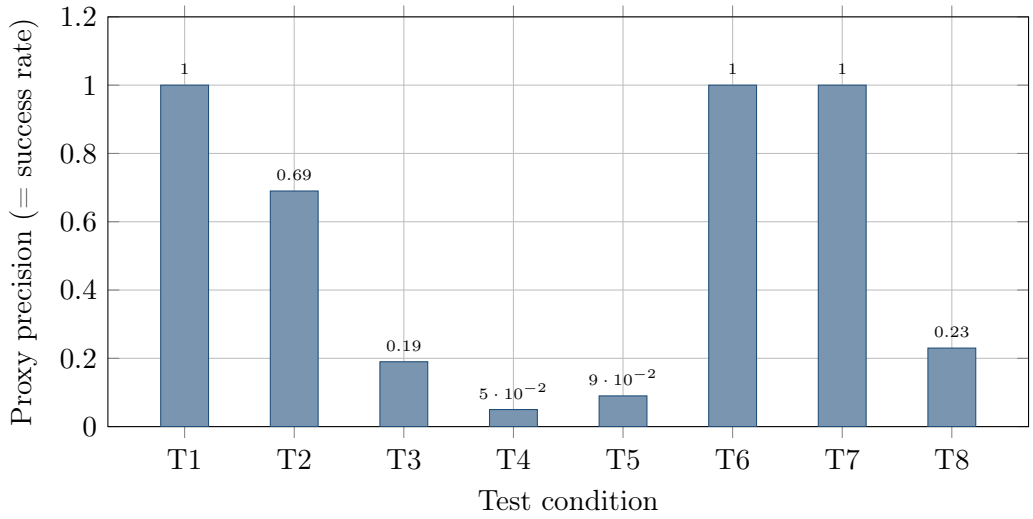


Figure 4.6: Proxy precision operating curve across all eight stress-test conditions. Recall is approximately 1.00 in every condition because the closure-validator fired in every episode; this curve therefore summarises the proxy operating profile across varying task difficulty. Conditions T1, T6, and T7 achieve precision 1.00. Spatial perturbation degrades precision sharply, with T4 ($\pm$10 cm) the worst-performing condition at 0.05. T5 recovers slightly due to edge-boundary artefacts at very large offsets.

A classical ROC curve plots the true-positive rate against the false-positive rate at varying decision thresholds. The closure-validator is a fixed binary detector with no adjustable threshold: in the tested regime it fires on every episode, which eliminates true-negative events and causes a conventional ROC curve to degenerate to a single operating point rather than a smooth trade-off curve. Figure 4.6 provides the closest equivalent: it characterises how the system's positive-predictive value changes across conditions of increasing perturbation severity. The visual message is the same as a degrading ROC curve, where a classifier's discrimination ability falls as the task becomes harder, but the degradation is expressed in the precision dimension rather than the TPR-FPR space. This is an inherent consequence of a validator that never withholds its trigger; improving specificity would require a stricter, threshold-based closure criterion calibrated against labelled grasp outcomes, which is identified as future work in Chapter 5.

The second limitation is spatial robustness. The final Condition B policy solved the fixed layout but failed under moderate and large cube-spawn perturbations. This makes the policy position-specific and limits claims about general manipulation.

The third limitation is statistical replication. The work uses one completed training run per condition. The stress test adds inference-time robustness evidence, but it does not solve seed variance.

The fourth limitation is evaluation scope. There is no physical robot deployment, no sim-to-real transfer, no tactile sensing, no visual distractor evaluation, and no multi-task benchmark. These exclusions are consistent with the methodology, but they narrow the claims that can be made.

4.15 Chapter Summary

This chapter presented the A/B training comparison and the eight-condition robustness evaluation. Condition A remained at 0% target-region success, showing that BC-only imitation was insufficient for the implemented contact-rich task. Condition B reached approximately 95% target-region success after PPO refinement and achieved 100% success in the fixed-layout T1 inference baseline, showing that reward-based refinement made the demonstration-assisted policy reliable in the trained scene.

The stress test added the missing robustness evidence. The final Condition B policy remained successful under observation noise alone, but spatial perturbation caused a severe performance drop: 69% success at ± 3 cm, 19% at ± 6 cm, 5% at ± 10 cm, 9% at ± 15 cm, and 23% under the combined ± 6 cm plus low-noise condition. This shows that the learned behaviour is robust to the tested observation noise but brittle under object-placement variation.

The final conclusion is therefore deliberately qualified. The dissertation demonstrates a functional low-cost R2S pipeline in which captured human-motion demonstrations can seed a simulated arm-hand policy and PPO can refine it to strong fixed-layout target-region performance. It does not demonstrate full release-based pick-and-place, physical robot transfer, multi-seed statistical generalisation, or broad spatial robustness. This critical boundary is the main difference between a visual proof of concept and a defensible applied research result.

Chapter 5

Conclusions and Recommendations

5.1 Chapter Overview

This chapter concludes the dissertation by evaluating the extent to which the research aim, research questions, and hypothesis were achieved. The study investigated whether captured human-motion demonstrations could be transferred into a simulated dexterous manipulation pipeline and whether the resulting virtual arm-hand system could learn the task sequence of grasping, lifting, transporting, and reaching a target region.

The final results support the hypothesis within a clearly bounded simulation context. Condition A, the BC-only baseline, remained at 0% target-region success during the submitted Condition A training run, up to approximately 1.206M logged steps. Condition B, the BC-assisted PPO refinement condition, reached approximately 95% target-region success after refinement and achieved 100 successful episodes out of 100 in the fixed-layout T1 inference baseline. This shows that the captured demonstrations provided useful task direction, but imitation alone was not sufficient; reward-based refinement was required for reliable fixed-layout target-region completion.

The conclusion must remain qualified. The system demonstrates simulated

grasp-and-transport-to-target behaviour under the implemented target-region success criterion. It does not demonstrate full release-based pick-and-place, physical robot deployment, tactile manipulation, multi-seed statistical generalisation, or broad spatial robustness. The stress-test evaluation is therefore central to the final conclusion: the policy solved the trained layout, remained robust to the tested observation noise, but failed sharply under cube-spawn perturbation.

5.2 Summary of Findings

The first major finding is that Behavioural Cloning alone was insufficient for the implemented contact-rich manipulation task. Condition A reached a closure-validator rate of 1.0 at convergence in the logged episode window, indicating that BC did learn the hand-closure component of the task. However, the dominant episode outcome was a drop rather than task completion: approximately 7 drops per 10-episode logging window with a timeout rate of 0%. BC reproduced grasp initiation but failed during post-grasp transport. This supports the literature-based concern that BC is vulnerable to covariate shift: once execution moves into states outside the demonstrated distribution, the policy has no reward-driven recovery mechanism to correct post-closure errors.

The second major finding is that BC-assisted PPO refinement produced a substantial improvement under the fixed task layout. Condition B used the same demonstrations as early assistance and then refined the policy through interaction with the reward function. The result shows that the demonstrations were valuable as a behavioural starting point, but that PPO refinement supplied the correction signal required for reliable task completion.

The third major finding is that the fixed-layout success rate does not imply broad generalisation. In the eight-condition robustness evaluation, the final Condition B policy achieved 100% success in the fixed baseline and 100% success under both tested observation-noise-only conditions. However, success dropped to 69% under ±3 cm cube-spawn perturbation, 19% under ±6 cm, 5% under ±10 cm, 9% under ±15 cm, and 23% under the combined ±6 cm plus low-noise condition. The dominant weakness was therefore spatial perturbation, not the tested observation noise.

The fourth finding is that reward design and validation design strongly affect interpretation. The final success criterion measures target-region entry, not stable release after hand opening. The simplified GraspValidator is treated as a closure-stability proxy rather than as ground-truth evidence of a physically valid grasp. For this reason, the reported success rates are interpreted as simulated grasp-and-transport-to-target results, not as proof of full autonomous pick-place-release behaviour.

5.3 Answers to the Research Questions

5.3.1 RQ1: Can a virtual arm and hand be trained within a simulation environment with R2S learning using captured human motion to accomplish basic grasping, lifting, transporting, and target-region completion?

The answer is condition-dependent. Under BC-only learning, the virtual arm and hand did not complete the implemented manipulation sequence. Condition A reached a closure-validator rate of 1.0 in the logged episode window, showing that BC learned hand closure, but drops dominated the episode outcomes and target-region success

remained at 0%. The failure was therefore not an absence of learning; it was a failure to sustain the grasp through the post-closure transport phase. Under BC-assisted PPO refinement, the virtual arm and hand did complete the fixed-layout target-region task. Condition B reached approximately 95% target-region success after refinement and achieved 100% success in the T1 fixed-layout inference baseline.

RQ1 is therefore supported for the BC-assisted PPO condition under the implemented target-region criterion. It is not supported for imitation alone, and it is not evidence of full release-stable pick-and-place.

5.3.2 RQ2: To what extent can the virtual arm and hand reproduce the task-level structure of the demonstrations and manipulate objects within the simulator?

Condition B reproduced the task-level structure of the demonstrations at the level evaluated in this dissertation: approach, hand closure, object movement, transport, and target-region completion. It also manipulated the cube inside the simulator because success required the object, not merely the end-effector, to enter the target-region radius.

However, the stress test shows that this task-level reproduction was strongly tied to the trained object placement. The policy remained successful when the cube started in the familiar position, but spatial offsets degraded performance sharply. The defensible conclusion is therefore that the policy reproduced the demonstrated task structure in the fixed scene, but did not learn robust object-location generalisation.

5.3.3 RQ3: How efficient is the learned system with respect to performance, training time, and real-time or near-real-time use within the environment?

The results separate runtime efficiency from training efficiency. Once trained, the Condition B policy executed quickly inside the Unity decision loop. Successful episodes completed in less than one second in the measured inference cases, and observed Unity behaviour supported near-real-time simulated control. This runtime claim is descriptive rather than a hardware-independent latency benchmark.

Training efficiency was weaker. Condition A failed during the submitted BC-only training run, while Condition B required extended PPO refinement before reliable fixed-layout success was achieved. The system is therefore efficient at inference once trained, but the training process remains computationally expensive and offline.

5.4 Hypothesis Evaluation

The research hypothesis stated that a virtual arm and hand trained through R2S learning from captured human-motion data would be able to execute actions related to grasping, lifting, transporting, and target-region completion within the virtual environment, while operating at a speed sufficient for real-time or near-real-time simulated control.

The completed results support this hypothesis for Condition B within the implemented simulation boundary. BC-assisted PPO refinement produced reliable fixed-layout target-region performance and the final policy operated as an inference controller inside Unity. The null hypothesis is therefore not supported for the com-

pleted Condition B fixed-layout experiment.

This is not a formal statistical rejection of the null hypothesis across independent training seeds. Only one completed training run per condition was used, and the A/B comparison was not a multi-seed inferential test. The hypothesis is supported descriptively for the implemented system, not proven as a general robotics claim across embodiments, random seeds, physical hardware, or unseen environments.

5.5 Contribution of the Dissertation

This dissertation makes a bounded empirical and methodological contribution. It demonstrates a working low-cost R2S pipeline in which captured human hand-motion intent is recorded as demonstrations, transferred into a simulated dexterous arm-hand environment, and used to support reinforcement learning refinement.

The first contribution is the controlled A/B comparison. Using the same demonstrations, task scene, observation space, and action space, the BC-only condition failed while the BC-assisted PPO condition solved the fixed-layout target-region task. This provides applied evidence that reward-based refinement was necessary in the implemented contact-rich manipulation setup.

The second contribution is the robustness evaluation. The eight-condition stress test prevents the fixed-layout success rate from being overclaimed. It shows that the final policy was robust to the tested additive observation noise but brittle under spatial cube-spawn perturbation. This makes the contribution more credible because the work identifies not only where the system succeeds, but also where it fails.

The third contribution is the critical qualification of the success metric.

The dissertation explicitly distinguishes target-region grasp-and-transport from full release-stable pick-and-place. It also treats the GraspValidator output as a proxy diagnostic signal rather than as ground-truth grasp evidence. This prevents the implementation from being presented as more general or more physically complete than it is.

5.6 Limitations

The main limitation is the success criterion. Target-region entry does not prove controlled release. The cube is considered successful when it enters the placement radius, not when it is released, remains still, and stays within the target region after hand opening. The result must therefore be interpreted as grasp-and-transport-to-target behaviour rather than full pick-place-release.

The second limitation is spatial robustness. The fixed-layout task was solved, but the stress test showed major performance loss under cube-spawn perturbation. This means the policy learned a position-specific manipulation behaviour rather than a broadly general object-relative strategy.

The third limitation is the simplified grasp validation. The GraspValidator was reduced to a closure-stability proxy because more physically grounded trigger checks were unreliable in the completed prototype. No independent manually labelled grasp dataset was collected; therefore, true positives, true negatives, false positives, false negatives, precision, recall, and F1 score are not reported as formal classifier metrics.

The fourth limitation is statistical replication. The dissertation reports one completed training run per condition. The stress test evaluates one trained Condition B policy across 800 inference episodes, but this does not replace multiple inde-

pendently trained policies with different random seeds.

The fifth limitation is evaluation scope. The study does not include physical robot deployment, tactile sensing, visual distractor testing, object-shape variation, random target positions, multi-object manipulation, or sim-to-real transfer. These omissions narrow the claim to the implemented Unity simulation environment.

5.7 Recommendations and Future Work

The first recommendation is to add a release-stability success criterion. Future versions should require the cube to remain inside the target radius at low velocity for a fixed stability window after hand opening. This would reduce reward-hacking opportunities and move the task closer to genuine pick-place-release behaviour.

The second recommendation is to improve spatial robustness through training-time variation. The stress test shows that spatial offset is the main weakness, so future work should use cube-spawn randomisation, curriculum learning from small to larger offsets, and object-relative observations or rewards. The goal should be to make the policy respond to the cube's current position rather than memorising a fixed start layout.

The third recommendation is to repeat the experiment across multiple random seeds. At least three independent runs per condition would allow mean performance, variance, confidence intervals, and appropriate non-parametric tests to be reported. This would convert the current descriptive comparison into a stronger statistical evaluation.

The fourth recommendation is to improve grasp validation. Future work should collect manually labelled grasp outcomes or implement more reliable contact, lift, and

post-release stability checks. This would allow formal true-positive, false-positive, true-negative, and false-negative analysis of the validator rather than proxy-only interpretation.

The fifth recommendation is to expand the evaluation matrix after the fixed task is robust. The next stages should include visual distractors, random target positions, object-shape variation, object-category changes, and eventually physical robot transfer. These should be treated as separate evaluation stages, not assumed from the current fixed-scene result.

5.8 Final Remarks

The dissertation achieved its core objective within the implemented simulation boundary. It built a working vision-based R2S dexterous manipulation prototype and showed that BC-only imitation learned hand closure but was insufficient for post-grasp transport, while BC-assisted PPO refinement resolved that transport phase and produced strong fixed-layout target-region performance. The result matches the pattern identified in the reviewed literature: demonstrations can provide useful task direction, but reward-based refinement is needed to overcome contact-rich execution errors and covariate shift.

The most important final insight is not simply that Condition B succeeded. It is that Condition B succeeded under the fixed layout, remained robust to tested observation noise, and failed under spatial object-placement variation. This distinction prevents the work from being reduced to a visual proof of concept. The dissertation demonstrates a functional pipeline, identifies the limits of its success, and shows which technical weaknesses must be solved next.

The final claim is therefore precise: the system learned simulated grasp-and-transport-to-target behaviour from captured human-motion demonstrations and BC-assisted PPO refinement. It did not prove full physical robotic dexterity, controlled release, multi-seed statistical generalisation, or broad spatial robustness. Within that bounded claim, the dissertation presents a defensible applied research result.

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Appendix A

Implementation and Reproducibility Artefacts

A.1 Project Artefacts

The implementation artefacts for this dissertation are submitted alongside the dissertation as a compressed project folder. The main files relevant to reproducibility are summarised in Table A.1.

Table A.1: Main repository artefacts supporting the dissertation.

Artefact	Purpose
`Dissertation-Project/`	Unity simulation project containing the KUKA iiwa7, SCHUNK SVH hand, task scene, reward logic, and capture interface.
`condition_a_bc.yaml`	ML-Agents configuration for Condition A: BC-only.
`condition_b_bc_ppo.yaml`	ML-Agents configuration for Condition B: BC-assisted PPO refinement.
`Demos/`	Recorded demonstration files and `demo_stats.csv`.
`Docs/LaTeX/`	Final dissertation source files.

A.2 Agent Interface Summary

Table A.2: Final ML-Agents interface used by the implemented prototype.

Item	Value
Behaviour name	`PickPlace`
Observation vector size	27
Continuous actions	13
Arm action dimensions	7 joint-delta actions
Hand action dimensions	6 values: thumb, index, middle, ring, pinky, and spread
Robot embodiment	KUKA iiwa7 arm with SCHUNK SVH five-finger hand

A.3 Demonstration Statistics Log

The file `demo_stats.csv` is generated by `DemoStatsLogger.cs`. It records one row per demonstration episode and stores the following fields:

- episode number and timestamp;
- success flag, duration, validated-grasp flag, lifted flag, and dropped flag;
- minimum distance to the target marker;
- time to grasp and time to lift;
- average thumb, index, middle, ring, pinky, spread, and wrist-roll values.

These data support interpretation of the demonstration distribution. They are not treated as trained-policy evaluation results.

A.4 Result Artefact Register

Only artefact-level summaries are provided here; the numerical interpretation of these files is reported in Chapter 4.

The result artefacts supporting Chapter 4 are listed in Table A.3.

Table A.3: Result artefacts retained with the submitted project for reproducibility checking.

Artefact	Purpose
TensorBoard event files	Scalar training curves for reward, success rate, time-to-success, and episode length.
`Assets/_Project/Models/PickPlace_ConditionB.onnx`	Exported trained Condition B policy artefact retained for reproducibility checking; not used for physical robot deployment.
`results/`	Per-condition stress-test episode logs (`stress_test_results.csv`).
`demo_stats.csv`	Demonstration recording summary: per-episode timing, outcome flags, and hand-signal averages.

Appendix B

Literature Review Support Matrix

B.1 Peripheral Literature Strands

Chapter 2 focuses on literature directly relevant to the implemented R2S, BC, and PPO comparison. Table B.1 records related strands that informed the background but were not central to the implemented experiment.

Table B.1: Peripheral literature strands and reason for appendix placement.

Strand	Reason for appendix placement
Bimanual manipulation	Requires two coordinated hands and changes the control problem beyond the single-arm, single-hand scope.
Tactile-informed manipulation	Depends on sensing modalities not implemented in the prototype.
Language-conditioned manipulation	Adds semantic task conditioning outside the fixed manipulation task.
Generated-video or demo-free approaches	Relevant to future work, but the present study uses captured human demonstrations.
Articulated tool use	Focuses on object articulation rather than cube grasp-and-transport.

B.2 Reviewed Study Matrix

Table B.2: Extended reviewed-study matrix supporting Chapter 2.

Study	Capture / supervision	Learning pattern	Relevance to dissertation	Boundary
DIME [8]	Teleoperation demonstrations	Imitation with policy improvement	Anchor evidence for BC fragility in dexterous manipulation.	Different task framework and hardware setup.
RialTo [7]	Real demonstrations aligned to simulation	Real-to-sim-to-real refinement	Strong R2S reference and BC-to-RL improvement pattern.	Includes physical transfer beyond this simulation-only study.
DexTransfer [21]	Minimal MoCap demonstrations	Few-shot dexterous grasp learning with refinement	Supports small-demonstration learning argument.	Uses MoCap and real-world dexterous grasping.
DexMV [9]	Human task video	Video-derived imitation with improvement	Supports vision-derived supervision and BC fragility argument.	Uses offline video rather than live capture.
DexCap [10]	Portable MoCap system	Demonstration collection and retargeting	High-fidelity capture baseline.	More instrumented than the present capture interface.
AnyTeleop [16]	Vision-based teleoperation	Cross-embodiment teleoperation	Relevant to embodiment mapping and retargeting.	More complex teleoperation system.
Open Teach [17]	Low-cost teleoperation rig	Demonstration collection and robot control	Accessibility and low-cost capture reference.	Uses a separate teleoperation system rather than this Unity R2S workflow.